Q-CRAFT Explorer Companion Guide

A course in climate-aware fiscal projection

Teal Insights & NatureFinance

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Preface

This is a course. It accompanies **Q-CRAFT Explorer**, an open-source Python reimplementation of the IMF's Quantitative Climate Risk Assessment Fiscal Tool, and it is written for economists at ministries of finance, IMF staff working on capacity development, and development practitioners who use fiscal projection tools in their work.

It has two objectives, and they are equally weighted: you should be able to **run** the tool, and to **understand what you are doing** when you run it. Those are different skills and the second one is the harder to fake in a meeting.

Q-CRAFT Explorer is developed by Teal Insights and NatureFinance. The code is open source (MIT license).

How the course is organised

Seven modules, each built around something you can do at the end of it. The worked country throughout is Uganda, which is also the tool's golden-master verification country, so every example runs on data you can check.

Module	What it gets you
Module 0	The capstone stated up front, a self-assessment, and the path that fits your background
Module 1	A full Uganda projection in ten minutes, and the map: seven modules, one equation
Module 2	The debt identity rebuilt from words. Skippable if you already use it.
Module 3	Every parameter set and defended, with the rationale written down
Module 4	Uganda end to end, from assumptions to a Fiscal Risk Statement paragraph
Module 5	What the tool cannot tell you, and which questions belong to a different tool
Module 6	The capstone brief and rubric, and a look back at where you started

Start at Module 0. It takes about ten minutes and it will tell you which of the later modules you can skip.

The Glossary defines key terms and the References section gives annotated pointers to the primary sources underlying the methodology.

There is also an appendix on co-design and the SovTech vision. It sits outside the learning path on purpose: it is about the project behind the tool rather than about using it, and nothing in the capstone depends on it. Read it if you want to shape what V2 does.

What this course defers to

This is an educational companion, not a replacement for the IMF's User Guide (Tim and Rahman, 2024), which remains the authoritative methodology reference. Where this guide explains a concept, it cites the relevant User Guide section so you can go deeper.

> **This is an initial version**
>
> Q-CRAFT Explorer is a starting point, not a finished product. We want to co-design the next version with you. See the co-design appendix for how to get involved.

> **Try the App**
>
> **Open Q-CRAFT Explorer** to follow along as you read. The source code is on GitHub.

Chapter 1

Start here: what you will walk into the PS's office with

What this module does:
- States the deliverable you will build, so you can judge every later module against it
- Calibrates what you already know, in about ten minutes
- Routes you to the path that fits, so nobody sits through material they do not need

1.1 The deliverable, stated up front

Uganda's Fiscal Risk Statement for FY 2024/25 has a section called "Climate Change Fiscal Risks." It runs from page 13 to page 17. It reports a baseline fiscal path to 2099, five climate scenarios, the GDP loss under each, and this finding: in the High, Hot and Vulnerable scenarios, public debt passes the 50 percent of GDP fiscal rule ceiling. The source line under the figures reads "QCRAFT (2023)."

That section came out of a five-day workshop the IMF's Fiscal Affairs Department ran for MoFPED staff in September 2023, written up as *Uganda: PFM Climate Assessment* (IMF, 2024). The tool that produced it is the tool this course teaches.

By the end of this course you will produce two things:

1. **An export packet.** The parameters you chose, a written rationale for each judgment call, and the polished charts and CSV that Q-CRAFT Explorer exports.
2. **A two-paragraph draft** in the register of Uganda's Fiscal Risk Statement, suitable for handing to the Macroeconomic Policy Department.

That packet is the capstone. It is what you walk into the Permanent Secretary's office with. Chapter 7 contains the brief and the rubric it will be marked against. Every module in between either feeds that packet or tells you when not to trust it.

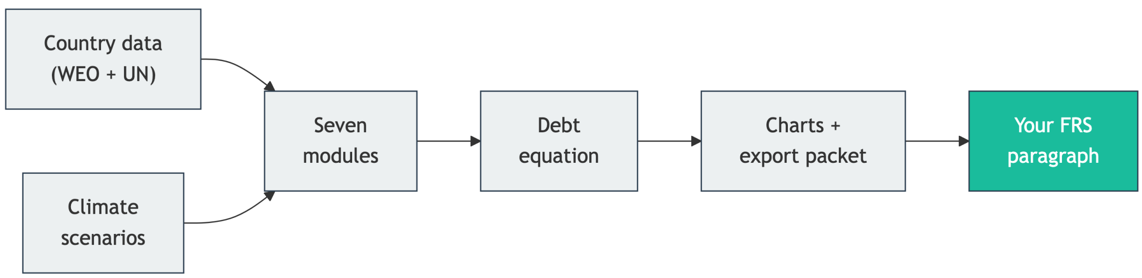
What this course is not

It is not a replacement for the IMF's User Guide (Tim and Rahman, 2024), which remains the authoritative methodology reference. Where this guide explains a concept, it cites the User Guide section so you can go deeper.

1.2 By the end of this module you can

- **Locate** your own starting knowledge against three anchored descriptions of practice
- **Diagnose** which of three concept-inventory items you answered from intuition rather than from the debt identity
- **Select** the path (A, B or C) that matches your background, and justify the choice
- **Describe** the capstone deliverable in one sentence to a colleague who has not taken the course

1.3 Where you are in the course



The course concept map. This module fixes the destination on the right.

This map reappears at the top of every module with the current node highlighted. Right now only the destination is lit, because that is the only part you need to hold.

1.4 Self-assessment: which of these describes your Monday?

Pick the one closest to true. There is no scoring and no one sees it. The point is to route you, and to give you something to compare against in Chapter 7.

On long-term fiscal projection:

	Description
A1	I have not worked with a debt sustainability analysis. I know what the debt-to-GDP ratio is and roughly why it matters.
A2	I have read DSAs and can follow the tables. I could not build the projection myself without help.
A3	I have produced or reviewed a DSA for a country. I could set up a baseline projection this week if asked.

On climate-fiscal analysis:

	Description
B1	I work on climate policy or climate finance. I have not connected it to a debt projection.
B2	I have seen climate scenarios used in fiscal analysis and can describe what they are for.
B3	I have run or interpreted a climate-scenario fiscal projection for a country.

On the tool itself:

	Description
C1	I have never opened Q-CRAFT, in Excel or on the web.
C2	I have opened it and clicked around. I could not defend the parameter settings I used.
C3	I have run Q-CRAFT for a country and explained the output to someone else.

Write your three answers down. You will be asked for them again at the end.

1.5 Three questions before you start

These are not a test. They are a check on which intuitions you are carrying in. Answer each one, commit to it in writing, then open the answer.

! DRAFT FOR TEAL: concept-inventory question 1 (interest-growth differential)

Question. A country's government runs a primary balance of exactly zero: revenue equals non-interest spending, to the shilling. The nominal interest rate on its debt is 9 percent. Nominal GDP growth is 6 percent. Over the next year, the debt-to-GDP ratio will:

(a) Fall, because the government is not adding to its debt.
(b) Stay flat, because a zero primary balance is by definition debt-stabilizing.
(c) Rise, even though the government borrowed nothing new for programmes.

Answer

(c) Rise. Debt grows at the interest rate; the economy it is measured against grows more slowly. The ratio climbs by roughly (9 - 6) / 1.06, about 2.8 percent of the starting ratio, in the first year. A country starting at 50 percent of GDP ends the year near 51.4 percent having changed no policy at all.

If you picked (b), you are carrying the most common and most expensive misconception in this material: that a zero primary balance stabilizes debt. It stabilizes debt only when the interest rate equals growth. Every other time, the interest-growth differential is doing work you have to pay for. Chapter 3 is built around this.

If you picked (a), you may be thinking of the deficit rather than the ratio. The stock of debt is flat in shillings; the ratio is not, because the denominator moved.

! DRAFT FOR TEAL: concept-inventory question 2 (expenditure rigidity)

Question. In Q-CRAFT Explorer you set expenditure rigidity to 1.0 and run a climate scenario. What have you assumed?

(a) Spending falls in step with GDP, holding the expenditure-to-GDP ratio at its baseline level.

(b) Spending stays at its baseline level in shillings even as GDP falls, so the whole revenue loss lands on the deficit.

(c) Spending is capped by the fiscal rule, whatever happens to GDP.

Answer

(b). Rigidity 1.0 means spending is sticky: the wage bill, pensions and existing commitments do not move when growth disappoints, so the entire climate-driven revenue shortfall becomes borrowing. It is the worst case, and it is the tool's default, deliberately.

If you picked (a), you have the scale inverted. That is rigidity 0.0. The inversion matters: it flips the sign of your headline risk number, and it is the single easiest way to publish a wrong figure from this tool.

If you picked (c), you have merged two separate controls. The fiscal rule and expenditure rigidity are different parameters and they act on different scenarios. Chapter 4 separates them.

! DRAFT FOR TEAL: concept-inventory question 3 (what the output is)

Question. Your Q-CRAFT run shows debt-to-GDP at 66 percent in 2099 under the Hot scenario against 47.5 percent in the baseline. A colleague asks what that means. Which answer is defensible?

(a) Uganda's debt will be about 66 percent of GDP in 2099 if the world warms on that path.

(b) Under a set of stated assumptions, climate damage of this magnitude would add roughly 19 percentage points of GDP to the debt ratio by 2099 relative to the same projection without climate damage.

(c) Climate change will cost Uganda 19 percent of GDP.

Answer

(b). The number that survives scrutiny is the *difference between two runs of the same model*, stated with its assumptions attached. Q-CRAFT is a scenario comparator, not a forecaster.

(a) treats a 2099 projection as a prediction. Nobody forecasts 2099. The tool's own User Guide says the results are not forecasts (pp. 5-6).

(b) converts a debt-ratio gap into a cost figure. It is not one. It is the additional borrowing implied by lower growth under stated behavioural assumptions, and it omits disasters, sea-level rise and adaptation spending entirely. Chapter 6 is about exactly this class of error.

The 47.5 and 66 percent figures are real: they come from the September 2023 IMF workshop with MoFPED staff, reported in the C-PIMA high-level summary (IMF, 2024).

1.6 Pick your path

The three paths cover the same tool. They differ in how much of the economics is rebuilt for you along the way.

Path	Choose it if	Route	Rough time
A. Guided	You answered mostly A1/B1/C1. Common for Climate Finance Unit staff who know the climate side and not the debt arithmetic.	M0 → M1 → **M2** → M3 → M4 → M5 → M6	5 to 6 hours
B. Standard	You answered mostly A2/B2/C1 or C2. Common for macro team staff: you know debt dynamics, the tool is new.	M0 → M1 → M3 → M4 → M5 → M6	3 to 4 hours
C. Fast	You answered A3 and C2 or C3. You have built projections and want the tool's specifics and the judgment calls.	M0 → M1 (early win only) → M3 self-checks → M4 → M5 → M6	2 hours

Chapter 3 exists for Path A. If you can already say what the interest-growth differential does to a debt ratio without looking it up, skip it. Forcing experienced staff through novice material makes the material harder, not easier.

Every module carries a fast-path marker near the top telling you the shortest useful route through it.

> **A note on time**
>
> Budget three to four times whatever you think this will take. That ratio is the standard finding on expert time estimates for other people's learning, and it holds here. The 10-minute Uganda run in Chapter 2 is genuinely 10 minutes. The rest is not.

1.7 Wrapper: what you should have now

- The capstone in one sentence: an export packet with documented rationale, plus a two-paragraph Fiscal Risk Statement draft.
- Three self-assessment answers written down for comparison in Chapter 7.
- Three concept-inventory answers, and a note on which ones you got from intuition rather than from the identity.
- A path.

On your desk this week: open Uganda's Fiscal Risk Statement FY 2024/25 to page 13 and read Section III. That is the genre you are writing into.

Chapter 2

One equation decides the debt path, and seven modules feed it

What you need to know:

- Q-CRAFT projects how climate change affects a country's debt trajectory through 2099
- Seven modules produce inputs to a single debt dynamics equation. Nothing else in the tool matters as much as that sentence.
- The Python Explorer reproduces the IMF's Excel workbook: baseline parity is exact for 147 of 147 tested countries

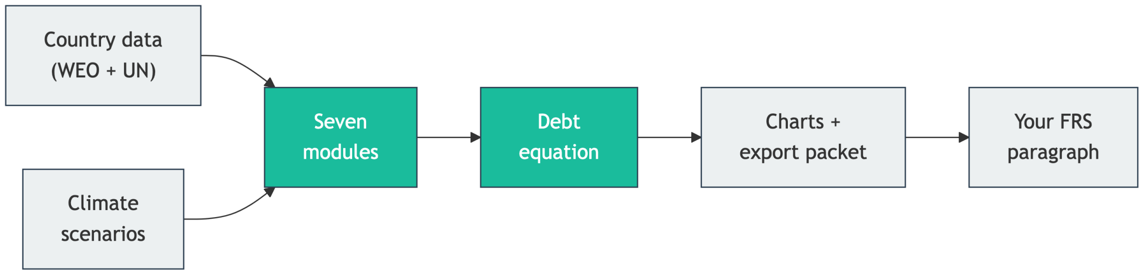
Fast path

Short on time? Do Section 2.4 (the Uganda run) and Section 2.6 (the module table and its self-check). Skip the rest and come back to it.

2.1 By the end of this module you can

- **Run** a full Uganda projection in Q-CRAFT Explorer and export the results
- **Name** the seven modules and **identify** which term of the debt equation each one moves
- **Locate** any parameter you are asked about in the correct module, without searching
- **Explain** to a colleague why a Python reimplementation of an Excel tool needs a parity test, and what "147 of 147" does and does not cover

2.2 Where you are in the course



This module covers the middle of the chain: the seven modules and the equation they feed.

2.3 Why a new platform for an existing tool

The IMF's Fiscal Affairs Department built Q-CRAFT as an Excel workbook. That decision made sense. Excel is universal. Every Ministry of Finance has it. Every economist knows how to use it. The Q-CRAFT workbook is well-structured, clearly documented, and covers 197 economies. The team that built it made a tool that works.

The challenge is what happens next.

A fiscal economist in a low-income country's Ministry of Finance opens the workbook. It has 19 sheets. The macroeconomic data is pre-loaded from the IMF's World Economic Outlook, but the WEO vintage may be a year old. The productivity growth assumption needs to reflect the country's structural outlook, but the workbook shows one default (1.2%) for all countries. Was that default used in last year's C-PIMA assessment, or did the visiting IMF team change it? The workbook doesn't record that history.

A year later, the next analyst opens the same file. There is no way to tell which assumptions were deliberate choices and which were left at defaults. There is no audit trail, no version history, no automated checks.

These are not failures of Excel. They are limitations of a format that was designed for calculation, not for workflow management, reproducibility, or collaborative analysis across institutions and time.

The practical cost is measured in capacity development budgets. Countries need technical assistance to set up and interpret these projections. The IMF sends economists to work alongside ministry staff, valuable work, but expensive to scale. If practitioners could start from a higher base, with a tool that guides them through parameter selection and records their analytical choices, that capacity development investment would stretch further.

Q-CRAFT Explorer puts the same proven economics on a modern platform. The calculation engine is a standalone Python package with extensive automated testing. The web interface loads bundled macroeconomic data (currently WEO October 2024 vintage) automatically, provides guidance at point of need, and exposes the key projection parameters in an interactive sidebar. The code is open source (MIT license): anyone can examine, verify, and extend it.

2.4 Run Uganda in ten minutes

Before any of the machinery is explained, produce the output. The explanation lands better on something you have already seen move.

> Zero to a Uganda projection
>
> **Open Q-CRAFT Explorer** and follow these steps:
> 1. **Select Uganda** from the sidebar dropdown. WEO macroeconomic data and UN population projections load automatically. Nothing to download, nothing to paste.
> 2. **Read the sidebar context.** It shows the latest WEO debt-to-GDP ratio and total population. Check them against what you know before going further.
> 3. **Open the Baseline tab.** This is Uganda's fiscal trajectory with no climate damage. Note the debt-to-GDP number at 2099.
> 4. **Open the Climate tab.** Six warming scenarios, their effect on GDP. Note where the scenarios separate from each other, and in which year.
> 5. **Open the Analysis tab.** Baseline and all climate scenarios on one debt chart. The vertical gap between the baseline line and the Hot Unadapted line is the number your Fiscal Risk Statement paragraph will be about.
> 6. **Open the Data tab and download both CSVs.** That is the first piece of your capstone export packet. Keep it.

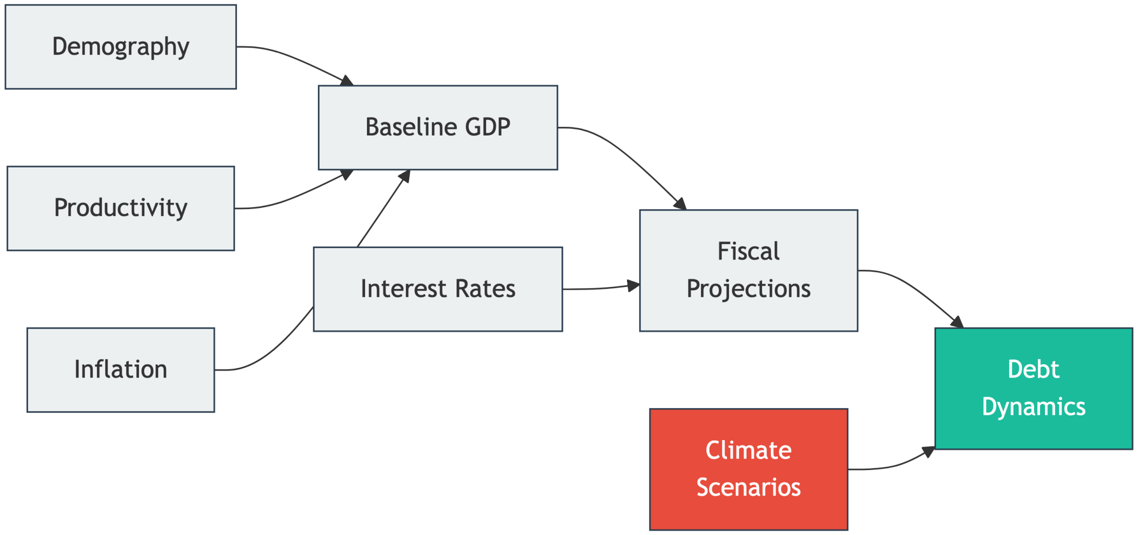
You have now done, with five clicks, the analysis that MoFPED staff produced in a five-day workshop with a visiting IMF team in September 2023. The workshop's finding, published in the C-PIMA high-level summary: GDP loss by end of century could pass 4 percentage points, with debt rising to 66 percent of GDP against 47.5 percent in the baseline (IMF, 2024).

Your numbers will not match theirs exactly. That workshop used the 2023 WEO vintage and its own parameter choices; the Explorer ships with WEO October 2024. Which vintage and which parameters were used is precisely the kind of thing that gets lost, and precisely what the export packet is for.

> SCREENSHOT-TODO
>
> Annotated screenshot of the Analysis tab for Uganda, with the baseline-to-Hot-Unadapted gap called out on the chart. Do not ship this module without it: the sentence "the vertical gap is the number your paragraph is about" is doing visual work with no visual.

2.5 What Q-CRAFT computes

Q-CRAFT projects long-term fiscal outcomes under different climate scenarios. It is a partial-equilibrium tool: it does not model feedback effects from fiscal policy changes on GDP growth or interest rates (User Guide, p. 5). It connects seven analytical modules, each feeding into a central debt dynamics equation.



Q-CRAFT's seven modules flow into the debt dynamics equation

2.5.1 The equation, in one sentence and then in symbols

Next year's debt ratio is this year's, grown by the interest rate, shrunk by economic growth, less whatever the government paid down.

That sentence is the whole model. The compact form is the sentence in shorthand:

$$d_t = d_{t-1} \times \frac{1 + r_t}{1 + g_t} - pb_t$$

Symbol	Meaning
d_t	Debt-to-GDP ratio in year t
r_t	Effective nominal interest rate on government debt
g_t	Nominal GDP growth rate
pb_t	Primary balance as a share of GDP (revenue minus non-interest spending)

When interest rates exceed growth ($r > g$), the debt ratio rises automatically: the government must run a primary surplus just to keep debt stable. When growth exceeds interest rates ($g > r$), debt dynamics are favourable and the ratio declines even with modest deficits. Chapter 3 works this through with numbers; if that paragraph was already obvious, skip it.

Climate change worsens debt dynamics through two indirect channels. It reduces GDP growth (g), and, when expenditure is rigid, it worsens the primary balance (pb) as revenue falls with GDP while spending stays elevated. There is no separate climate term in the equation. The climate impact is embedded in g and pb, which is why finding it in the output takes a comparison rather than a single number.

Every module in Q-CRAFT exists to produce an input to this equation.

2.6 How each module feeds the equation

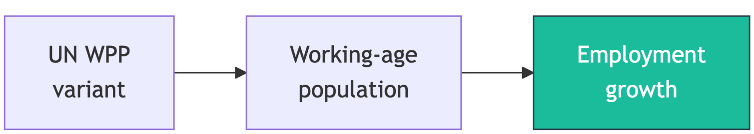
Seven modules, and each one moves exactly one part of the equation. The table is the thing to remember. The diagrams below it are for the module you happen to be arguing about.

Module	What it produces	Which term it moves	Do you control it in V1?
Demography	Working-age and total population to 2100	g (through employment)	Yes: Medium / High / Low variant
Productivity	GDP per employed person, growth path	g	Not in V1 (defaults match Excel)
Inflation	Price path converging to target	g (nominal)	Not in V1 (defaults match Excel)
Baseline GDP	Nominal GDP growth, combining the three above	g	Indirectly, through the above
Interest rates	Effective nominal rate on the debt stock	r	Not in V1 (defaults match Excel)
Fiscal projections	Revenue, primary expenditure, primary balance	pb	Yes: debt target, fiscal rule on/off
Climate scenarios	GDP damage by scenario, from 2030	g, and pb via rigidity	Yes: expenditure rigidity

Read the last column carefully. V1 exposes five controls; the other assumptions sit at the Excel tool's defaults. That is a deliberate scoping choice, not an oversight, and Chapter 6 says what it costs you.

> ⚠ WIDGET-TODO: seven-modules-to-one-equation
>
> Embed the seven-modules-to-one-equation intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The integration pass happens after this restructure and run 3 both land; this callout is the anchor point. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

2.6.1 Demography



Population by age group (children, working age, elderly) from UN World Population Prospects, in three variants: medium, high, and low. Working-age population growth (ages 15-64) drives employment growth after the WEO forecast horizon, which in turn drives potential GDP. Countries with shrinking working-age populations face slower growth and more challenging debt dynamics.

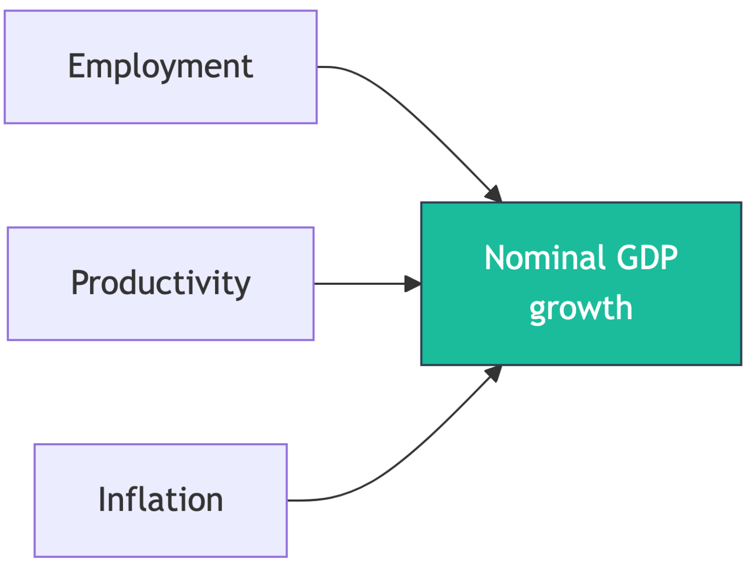
2.6.2 Productivity

GDP per employed person, from World Bank data. A starting growth rate and a long-run convergence rate, with Q-CRAFT transitioning between them along a logistic curve: growth shifts gradually, not abruptly. The productivity-relative-to-OECD trajectory serves as a realism check. Assumptions that imply a low-income country surpassing OECD productivity by 2099 warrant scrutiny.

2.6.3 Inflation

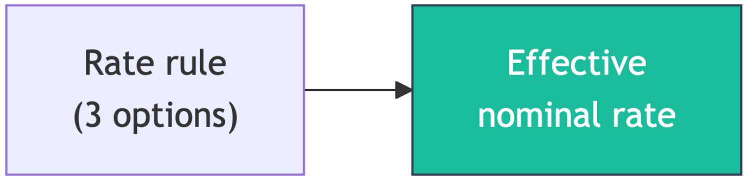
Same logistic convergence pattern. The starting rate should reflect the WEO projection for the final forecast year; the ending rate should reflect the long-run target, typically 2-3% for advanced economies and 3-6% for emerging and developing economies. For Uganda, the Bank of Uganda's 5% core inflation target is the reference point.

2.6.4 Baseline GDP



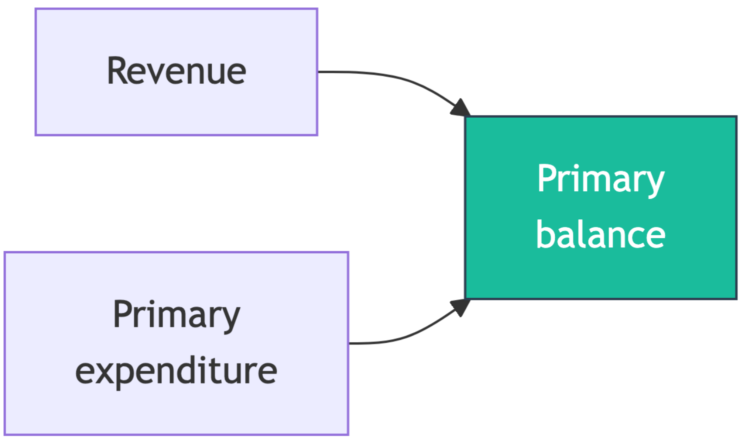
Nominal GDP growth decomposed into employment growth, productivity growth, and inflation. During the WEO forecast period (through 2029), Q-CRAFT uses the IMF's own projections. Beyond that horizon the model takes over using the assumptions set in the dashboard. This is the g in the equation.

2.6.5 Interest rates



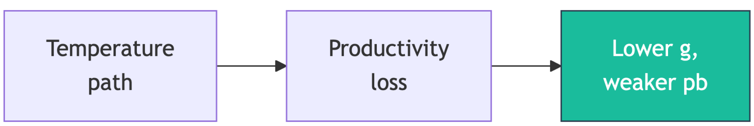
The cost of carrying the debt stock. Q-CRAFT offers three approaches: constant nominal rate, constant interest-growth differential, or constant real rate. Country expertise matters here. The choice should reflect concessional lending terms, sovereign risk premia, and the maturity structure of domestic financial markets. For a country moving from concessional to commercial borrowing, this assumption is doing more work than any other. This is the r in the equation.

2.6.6 Fiscal projections



Revenue grows with nominal GDP, maintaining a constant revenue-to-GDP ratio. Primary expenditure, the component excluding interest payments, grows multiplicatively with productivity, inflation, and total population, that is $(1 + \text{productivity growth}) \times (1 + \text{inflation}) \times (1 + \text{population growth})$, plus any fiscal rule adjustment. When enabled, the fiscal rule adjusts spending to steer debt toward a target ratio. This is the pb in the equation.

2.6.7 Climate scenarios



Empirical estimates of how temperature changes affect GDP, from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on the Kahn et al. (2021) methodology for long-term macroeconomic effects of climate change. Six scenarios span Paris-aligned (below 2°C warming) to hot-unadapted (high warming with slow policy response). Climate scenario impacts begin in 2030; before that year all scenarios match the baseline (User Guide, p. 19). Impacts flow through the productivity channel: reduced GDP growth means lower revenue, higher expenditure-to-GDP ratios, and accelerating debt.

The six climate scenarios

Scenario	Warming (wrt present)	Based on	Description
Paris-Aligned (1.5°C)	+0.7°C	SSP1-2.6	International commitments met; significant emission cuts
Moderate (2°C)	+1.6°C	SSP2-4.5	Current trends continue; no new aggressive mitigation
High (4°C+)	+2.5°C	SSP3-7.0	Countries scale back mitigation; fragmented response
Hot (3°C)	+3.5°C	SSP3-7.0 (90th pctile)	Same emissions as High but using the 90th percentile of temperature increases
Hot + Adapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot; countries adapt more quickly
Hot + Unadapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot; countries adapt very slowly

Warming values are IPCC best estimates for 2081-2100 relative to present (User Guide Table 1). Adding ~1.1°C for present-to-pre-industrial warming gives the more familiar above-pre-industrial framing: Paris <2°C, Moderate ~2.7°C, High ~3.6°C, Hot ~4.6°C.

Uganda's own Fiscal Risk Statement FY 2024/25 reports five of these, naming the last one "Vulnerable" rather than "Hot Unadapted". Same scenario, different label. Chapter 5 uses the Explorer's naming.

These scenarios are conservative in ways that matter for how you write up the results. Chapter 6 sets out what they leave out.

Self-check: where does it live?

A colleague sends you five questions in one email. For each, name the module you would open first. Commit to your five answers before opening the panel.

1. "Our fiscal rule ceiling is 50 percent of GDP. Does the model know that?"
2. "The projection assumes we keep growing at 6 percent forever. Does it?"
3. "Why does the debt path only start diverging across scenarios in 2030?"
4. "We are shifting from IDA credits to Eurobonds. Where does that show up?"
5. "Spending per person keeps rising in the projection. Who decided that?"

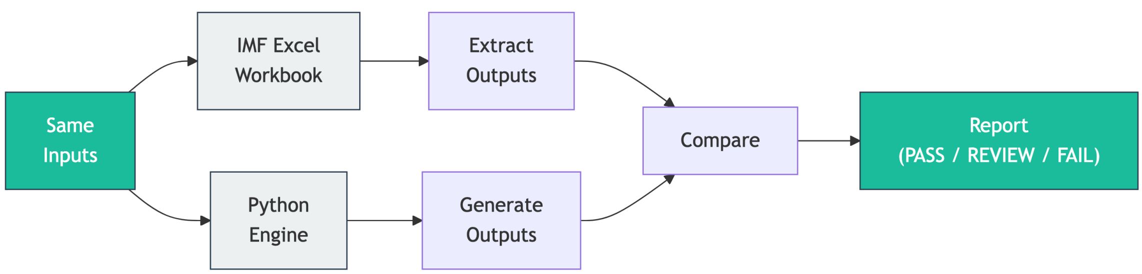
Answers

1. **Fiscal projections.** The debt target and the fiscal rule on/off switch are both there, and both are yours to set.
2. **Baseline GDP**, and behind it Productivity and Demography. Growth is built from employment plus productivity plus inflation, with a logistic transition, so it does not sit at the WEO rate forever.
3. **Climate scenarios.** Impacts start in 2030 by construction; before that every scenario equals the baseline (User Guide, p. 19).
4. **Interest rates.** The rate rule is where concessional-to-commercial transitions live. In V1 it sits at the Excel default, which is a limitation worth stating in your write-up.
5. **Fiscal projections.** Primary expenditure grows with productivity, inflation and total population, multiplicatively. Nobody decided it per person; it falls out of that formula.

If you missed 3, reread the climate module above. If you missed 5, the expenditure growth formula is the thing to hold: $(1 + a) \times (1 + b) \times (1 + c)$, never $a + b + c$.

2.7 Does it agree with the Excel tool?

A reimplementation is worth nothing if it quietly disagrees with the original. This is the one place in the course where the verification story appears, and it is here because you should decide now whether to trust the numbers.



The verification pipeline: same inputs, two engines, automated comparison

Identical inputs go through both the original Excel workbook and the Python engine. Every output cell for every projection year is compared. Think of it as an auditor reconciling two copies of the same ledger: if a single number differs, the test fails and we investigate.

Where that stands. Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation on debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP. An additional 25 parameter sensitivity combinations were tested and passed. Climate-scenario parity is confirmed for ratio metrics, meaning debt-to-GDP, revenue and expenditure as percent of GDP, across the tested countries and scenarios.

Those two sentences are the claim, and it is worth being precise about what the second one does not say. Climate-scenario agreement has been established for the ratios you will quote in a Fiscal Risk Statement. It has not been established across every level series in the workbook. If your analysis turns on a climate-scenario figure in levels rather than as a share of GDP, check it against the Excel tool yourself.

The test suite is public and runs on every change. You do not have to take the claim on trust; you can run it.

2.8 Wrapper: what you can now do

- You have run a full Uganda projection and exported the data. That file is capstone material.
- You can name the seven modules and say which term of the debt equation each one moves.
- You can say what the parity claim covers (baseline, 147 of 147, exact) and what it does not (climate levels).

Common error at this stage: treating the seven modules as seven things to learn. They are one equation with seven suppliers. If you can answer the “where does it live” self-check, you have the map.

On your desk this week: when someone hands you a fiscal projection, ask which of r, g and pb their headline number moves. It works on projections that have nothing to do with climate.

Chapter 3

You already know the debt equation

What you need to know:

- This module rebuilds the debt identity from words, for people who have not had to use it
- Two things move the ratio: the gap between the interest rate and growth, and the primary balance
- Everything Q-CRAFT does to model climate is an attack on one of those two

Fast path

This is the Path A module. If you can already say what happens to a debt ratio when nominal growth falls below the effective interest rate, and why a zero primary balance does not stabilize debt, skip to Chapter 4. Nothing later depends on reading this.

3.1 Warm-up

Two questions from Chapter 2, from memory:

1. Which term of the debt equation does the Demography module move?
2. In which year do the climate scenarios start to differ from the baseline?

(Answers: g, through employment growth. 2030.)

3.2 The hook: the question your minister actually asks

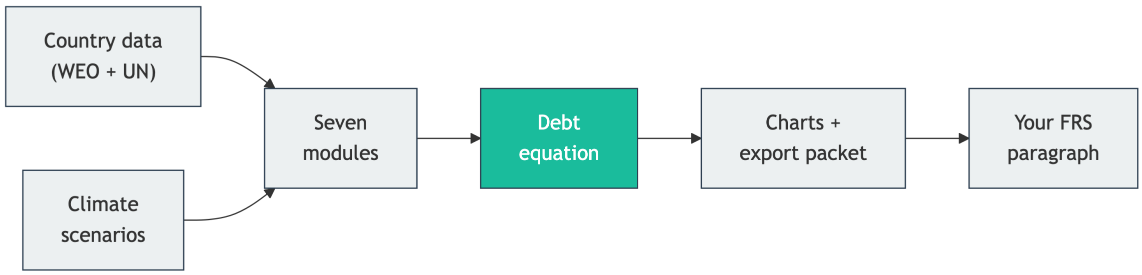
The question that arrives from above is rarely "what is the debt ratio in 2099." It is some version of: *if growth disappoints, how much trouble are we in, and how fast?*

That question has an exact answer, and it comes from one line of arithmetic that has been in every debt sustainability analysis you have ever read. This module is that line, taken slowly.

3.3 By the end of this module you can

- **Derive** the debt-ratio identity in words, without notation, and check that the notation matches
- **Predict** the direction and rough size of the debt-ratio change given r, g and the primary balance
- **Compute** the debt-stabilizing primary balance for a country from three numbers
- **Explain** why climate damage shows up in a debt projection with no climate term in the equation

3.4 Where you are in the course



This module is the equation node, on its own.

3.5 Predict first

Before any arithmetic. Write down your answers; you will check them at the end of the module.

1. A country's debt is 50 percent of GDP. It runs a primary balance of zero every year. Its effective interest rate is 9 percent, nominal growth 6 percent. Where is the ratio in ten years: below 50, near 50, or above 60?
2. Same country, but nominal growth is 11 percent. Same question.
3. Same country at 9 percent interest and 6 percent growth. What primary balance, as a share of GDP, would hold the ratio at exactly 50?

! DRAFT FOR TEAL: building the identity from words

Start with the stock, not the ratio. Government debt is a stock of shillings. Next year it is this year's stock, plus interest owed on it, plus any new borrowing to cover the gap between what the government spends on programmes and what it collects.

Debt next year = debt this year + interest on it + (non-interest spending − revenue)

That last bracket is the primary deficit. Flip its sign and it is the primary balance. Nothing controversial has happened yet.

Now divide by GDP, because that is how debt is judged. A debt ratio has a numerator that grows at the interest rate and a denominator that grows at the rate of nominal GDP. Two things are racing. The ratio moves according to which one is winning:

Debt ratio next year = debt ratio this year × (1 + interest rate) ÷ (1 + growth rate) − primary balance

Read the middle term as a scoreboard. If interest beats growth, the fraction is bigger than one and the ratio is amplified every single year, whether or not anyone borrowed a shilling for a new road. If growth beats interest, the fraction is less than one and the ratio deflates on its own.

That is the equation. In shorthand:

$$d_t = d_{t-1} \times \frac{1 + r_t}{1 + g_t} - pb_t$$

The shape is worth naming, because you will recognise it elsewhere: a stock carried forward, amplified by a ratio of two competing forces, minus a flow. Compound interest with a headwind and a repayment.

The one sentence to keep. Next year's debt ratio is this year's, grown by interest, shrunk by growth, less what you paid down. If you can say that without looking, you own the equation.

3.6 The scoreboard term: r minus g

! DRAFT FOR TEAL: what the interest-growth differential does

The amplifier $\frac{1+r}{1+g}$ is usually reported as a single number, the interest-growth differential $(r-g)/(1+g)$. Its sign is the most consequential fact about a country's debt position, and it is not a policy choice in the short run.

Sign trace. Take a debt ratio of 50 percent of GDP and a primary balance of exactly zero, so the second term drops out and only the scoreboard is left.

r	g	Ratio after 1 year	After 10 years	What is happening
9%	6%	51.4	66.1	Interest wins. The ratio climbs with no new programme borrowing.
6%	6%	50.0	50.0	Dead heat. A zero primary balance stabilizes debt, and only here.
6%	9%	48.6	37.8	Growth wins. The ratio falls while the government runs a balanced primary account.

Three percentage points of differential, held for a decade, is the difference between 66.1 and 37.8 percent of GDP. Nothing in that table is a policy decision about spending. It is arithmetic on two macro variables.

WIDGET-TODO: interest-growth differential

Embed the interest-growth differential intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The integration pass happens after this restructure and run 3 both land; this callout is the anchor point. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

Why this is the climate channel. Q-CRAFT does not add a climate term to the equation. It lowers g. Look at what that does to the first row: every year of slower growth makes the amplifier larger, and the effect compounds. A tenth of a percentage point off growth, sustained for seventy years, is not a rounding error in this equation. It is the mechanism.

Sensitivity, ranked. If you have time to interrogate one assumption in a Q-CRAFT run, interrogate the interest rate rule, because it sets r and is not exposed in V1. If you have time for two, add the productivity growth path, because it sets most of g. Everything else moves the answer less.

3.7 The flow term: the primary balance

! DRAFT FOR TEAL: the debt-stabilizing primary balance

The second term is the part the government controls. Revenue minus non-interest spending, as a share of GDP. Interest is excluded because it is not discretionary this year; it is the price of last year's decisions.

The question the term answers. Given where the scoreboard sits, what primary balance holds the ratio still? Set next year's ratio equal to this year's and solve:

$$pb^* = d \times \frac{r - g}{1 + g}$$

That is the debt-stabilizing primary balance, and it is a standard DSA quantity. It says: the worse your interest-growth differential and the higher your existing debt, the larger the surplus you need just to stand still.

Run it for the country in the table. Debt at 50 percent of GDP, r at 9 percent, g at 6 percent:

$$pb^* = 0.50 \times \frac{0.09 - 0.06}{1.06} = 0.0142$$

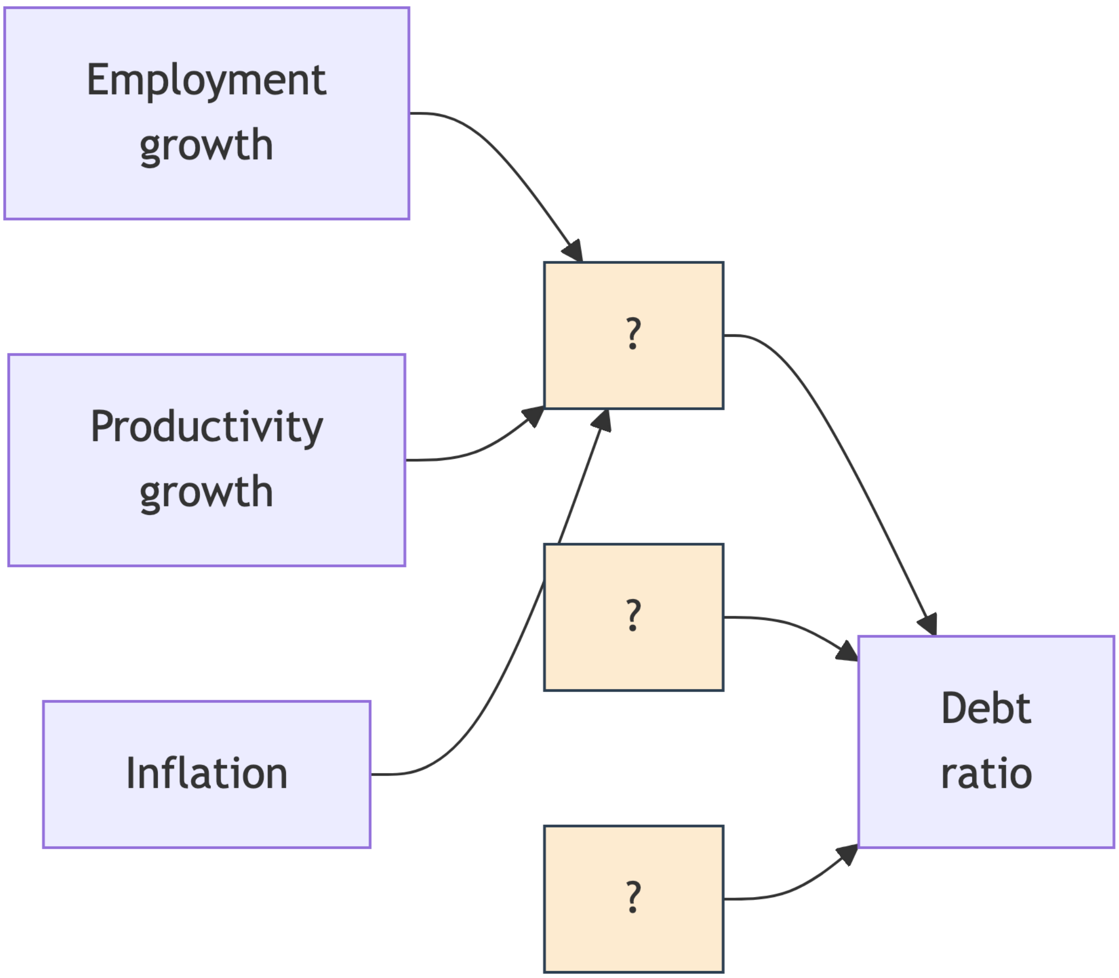
A primary surplus of about 1.4 percent of GDP, every year, to keep the ratio at 50. A government running a primary deficit of 1 percent in that position is not close to stable; it is 2.4 percentage points of GDP away from stable.

Perspective. For Uganda, 1.4 percent of GDP is on the order of the annual budget for a mid-sized ministry. That is what the arithmetic asks for before a single new priority is funded.

Now connect it to rigidity. Climate damage lowers g, which raises pb^*: you need a bigger surplus. It also lowers revenue, because revenue tracks nominal GDP. If spending does not fall to match, the actual primary balance moves the wrong way at the same time as the required one moves up. Those two effects push in the same direction, which is why the expenditure rigidity parameter in Chapter 4 has such a large effect on the answer.

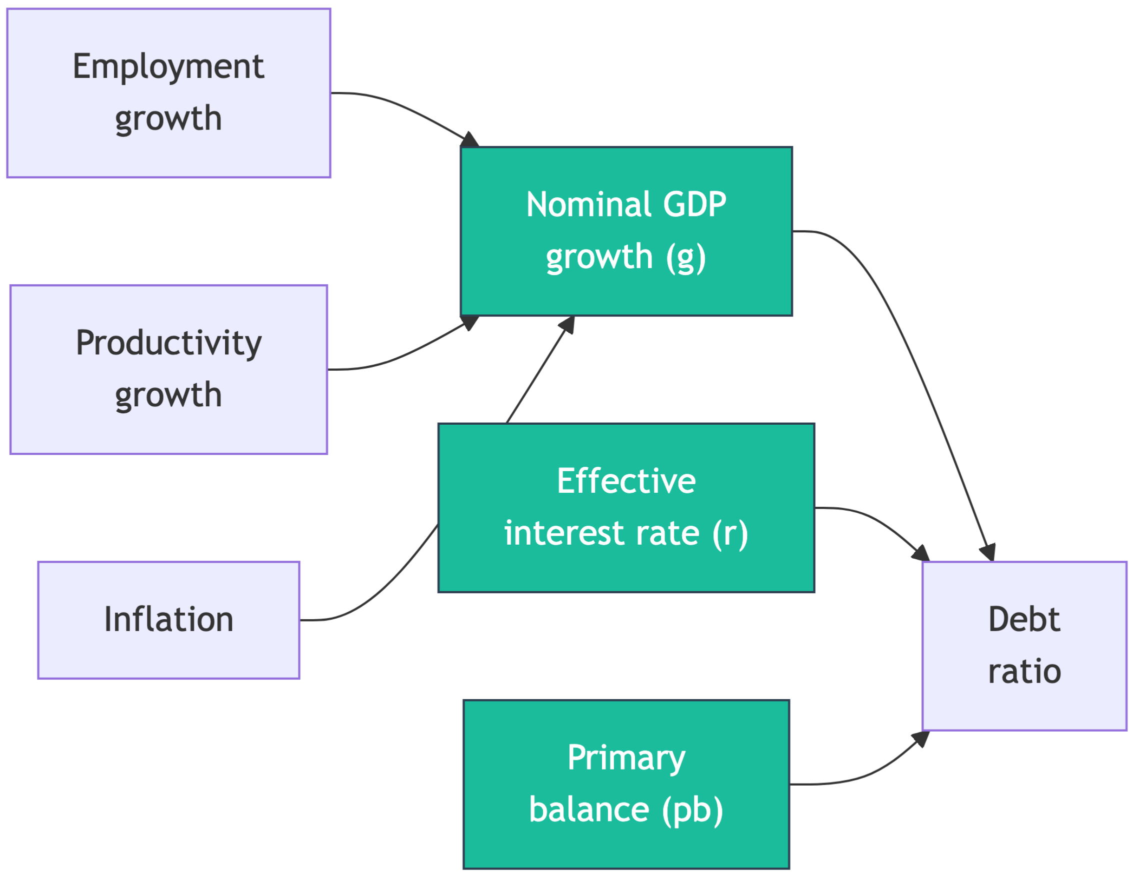
3.8 Complete the map

Below is the debt equation as a diagram with three nodes left blank. Fill them in before opening the answer. Constructing the map is the exercise; reading a finished one is not.



Fill in the three blanks.

The completed map



Three inputs, one output. If your version had climate as a fourth arrow into the debt ratio, redraw it: climate enters through g and through pb, never directly. That is the single most useful structural fact in this course, and it is why reading a climate result takes a comparison between two runs.

3.9 Predict, then run it

Now use the app. Predict before each click; the prediction is what makes the observation stick.

Predict, observe, explain

Setup. Open Q-CRAFT Explorer, select Uganda, fiscal rule **off**.
Predict. Write down: over 2030 to 2099, is Uganda's baseline debt ratio rising, flat, or falling? One sentence on why, naming r and g.
Observe. Open the Baseline tab. Read the debt-to-GDP trajectory. Then look at the Fiscal Balances chart and note whether the primary balance is in surplus or deficit.
Explain. Reconcile the two. If the primary balance is in deficit and the ratio is not exploding, the scoreboard is working in Uganda's favour: nominal growth is beating the effective interest rate, which is the normal position for a country still borrowing largely on concessional terms.

If your prediction assumed a deficit must mean rising debt, that is the misconception from Chapter 1 question 1, and now you have seen it fail on real data.

⚠ SCREENSHOT-TODO

Baseline tab for Uganda with the debt trajectory and the Fiscal Balances panel side by side, so the "deficit but stable ratio" reconciliation is visible in one image.

3.10 Self-check: three judgment calls

A colleague makes three claims

Your counterpart in another department sends these. For each, say whether you would agree, and what you would check.

1. "Our debt ratio fell last year, so the fiscal position improved."
2. "We should compare our primary deficit to the regional average to see if we are over-spending."
3. "Inflation is helping us: it inflates away the debt, so higher inflation is good for the ratio."

DRAFT FOR TEAL: self-check answers

1. **Not necessarily.** A ratio can fall on the denominator. Uganda's own Fiscal Risk Statement makes this point about FY2022/23: the debt ratio fell to 46.9 percent from 48.4, and the statement attributes it partly to nominal GDP rising on high inflation, warning the effect would subside as inflation fell. Check whether the stock of debt fell or the denominator moved.
2. **Wrong benchmark.** The primary deficit that is sustainable depends on your own r, g and starting debt, not on what neighbours do. The comparison that means something is your actual primary balance against your own debt-stabilizing primary balance, $d \times (r - g)/(1 + g)$. Two countries with identical deficits can be in opposite positions.
3. **Half right, and the half that is wrong is expensive.** Unexpected inflation does raise nominal g and deflate the ratio, once. But it feeds into the effective interest rate on new and floating-rate borrowing, so r follows g up with a lag, and the gain reverses. In Q-CRAFT specifically, revenue and primary expenditure both grow with inflation, so a permanently higher inflation assumption largely cancels out of the ratio. If you want to see this, it is a good use of the sensitivity runs in Chapter 4.

3.11 Check your predictions

Back to the three predictions from the start of the module.

1. **Above 60.** 66.1 percent after ten years. Zero primary balance is not stabilizing when $r > g$.
2. **Below 50.** 37.8 percent. The same zero primary balance, the opposite result, because the scoreboard flipped.
3. **A surplus of about 1.4 percent of GDP**, from $0.50 \times (0.09 - 0.06)/1.06$.

If you got all three, you did not need this module and Chapter 1 routed you wrong. If you got the third one only after seeing the formula, that is the expected outcome and you are ready for Chapter 4.

3.12 Wrapper: what you can now do

- You can build the debt identity from a sentence and check that the notation matches.
- You can compute a debt-stabilizing primary balance from debt, r and g.
- You can say where climate enters the equation, which is nowhere directly and everywhere through g and pb.

Common errors on this material: reading a falling ratio as fiscal improvement; assuming a balanced primary account stabilizes debt; drawing climate as a separate term.

One thing this module deliberately skipped: the baseline applies a floor of zero to the debt ratio and the climate scenarios do not. That asymmetry changes how you read the bottom of a fan chart, and it is covered in Chapter 6.

On your desk this week: compute the debt-stabilizing primary balance for Uganda from the current WEO numbers, and compare it to the actual. It is three numbers and one division, and it is the sentence that opens most credible fiscal risk write-ups.

Chapter 4

Every parameter is a judgment call you can defend

What you need to know:

- Q-CRAFT Explorer has five user-facing controls. Each one is a defensible choice, not a setting.
- Every parameter here gets the same treatment: what it is, why it matters, how to set it, predict-observe-explain in the app, and one judgment call to defend
- Every parameter also gets a line in your export packet saying why you chose what you chose. That line is what survives you leaving the ministry.

Fast path

Already run the tool? Do the five **Defend your choice** self-checks and the **Document it** blocks, and skip the What/Why prose. That is about 25 minutes and it is the part the capstone rubric marks.

4.1 Warm-up

From Chapter 2, without looking:

1. Which module owns the debt target and the fiscal rule?
2. Expenditure rigidity of 1.0 means spending is what?

(Answers: fiscal projections. Sticky, so the full revenue loss lands on the deficit.)

4.2 The hook: the assumption nobody wrote down

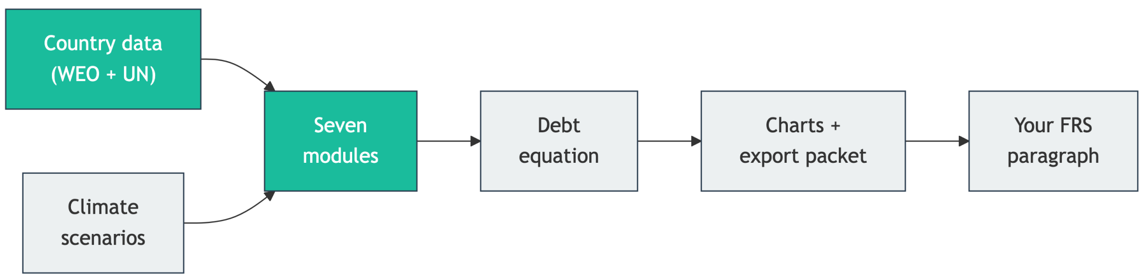
Reopen the vignette from Chapter 2. A year after the projection was built, nobody can say whether 1.2 percent productivity growth was a considered view about the country or a default nobody touched. The analysis is not wrong. It is unusable, because it cannot be defended.

Q-CRAFT Explorer cannot stop you leaving a default in place. What it can do is make the record: your parameters and your reasons travel with the output. This module is about earning the right to that record.

4.3 By the end of this module you can

- **Set** all five Explorer parameters for a country and **justify** each against country evidence
- **Predict** the direction and rough magnitude of each parameter's effect before you move the slider
- **Defend** an expenditure rigidity choice against a colleague who prefers a different one
- **Produce** the assumption-rationale section of the export packet, one entry per parameter

4.4 Where you are in the course



This module is the data and the parameters that drive the modules.

4.5 The four tabs, in one paragraph

You will be moving between the sidebar and the tabs constantly in this module, so here is the map of the screen. The **Baseline** tab is the no-climate-change projection. The **Climate** tab shows what warming does to GDP, scenario by scenario. The **Analysis** tab overlays baseline and climate debt paths on one chart. The **Data** tab gives you the grid and the CSV exports. Reading those charts properly is Chapter 5; for now you only need to know where to look when a parameter moves.

4.6 Setting the Parameters

Q-CRAFT Explorer has five user-facing parameters. The first (country selection) loads the data. The remaining four shape the projection.

Productivity, inflation and interest rate assumptions are not exposed in V1: they sit at defaults matched to the original Excel tool. That is a limitation with consequences, and Chapter 6 says what they are. It also means the four parameters below are carrying more of your analytical judgment than they would in the full workbook.

4.6.1 Country selection

What it is. A dropdown menu listing countries for which Q-CRAFT Explorer has complete data coverage across all required input datasets. Selecting a country loads macroeconomic data from the

IMF's World Economic Outlook (currently October 2024 vintage) and population projections from the UN World Population Prospects (2022 revision).

Why it matters. The country selection determines all historical data and WEO forecast-period projections. No manual data entry is needed: the tool loads real GDP, nominal GDP, revenue, expenditure, debt, interest rates, and demographic projections automatically.

How to set it. Choose the country you are analyzing. The sidebar displays key context after selection: the latest WEO debt-to-GDP ratio and total population. Use these as a quick sanity check: if these numbers do not match your understanding, investigate before proceeding.

> **Country coverage**
>
> Q-CRAFT Explorer currently covers 197 countries. Coverage is expanding as verification progresses. If your country is not listed, it may be added in a future update. See the IMF User Guide (Tim and Rahman, 2024), Section II.B for details on what data is loaded for each country.

> **Predict, observe, explain**
>
> **Predict.** Before you select Uganda, write down the debt-to-GDP ratio you expect the sidebar to report, from your own knowledge of the FY2023/24 position.
> **Observe.** Select Uganda and read the sidebar figure.
> **Explain.** If your number and the tool's differ by more than a point or two, work out which one is on a different basis before you go any further. Common causes: gross versus net debt, fiscal year versus calendar year, and the WEO vintage lagging the ministry's own outturn. Every one of those is a legitimate difference, and every one of them will be the first question you are asked about your results.

> **Defend your choice: country and vintage**
>
> A reviewer asks: "Is this the current debt number?" You are using WEO October 2024. Uganda's own FY2024/25 Fiscal Risk Statement reports public debt at 46.9 percent of GDP at June 2023 and projects 49.2 percent at June 2024. What do you say?
>
> **A defensible answer**
> Name the vintage, name the basis, and state which direction the difference runs. "This is WEO October 2024 on a calendar-year basis; the ministry's June 2024 figure is 49.2 percent on a fiscal-year basis. The starting point differs by under a point, and since the climate result is a difference between two runs from the same starting point, it is not sensitive to that gap." The last clause is the one that ends the conversation, and it is only available to you because you understand what the tool computes.

Document it

In your export packet, record: **data vintage** (WEO October 2024, UN WPP 2022), **starting debt ratio as loaded**, and **any known difference from your ministry's own figure, with the reason**.

4.6.2 Demography variant

What it is. The UN publishes population projections in three variants (Medium, High, and Low) reflecting different fertility assumptions. This parameter selects which variant Q-CRAFT uses for population and labor force projections through 2100.

Why it matters. Working-age population growth (ages 15-64) drives employment growth in Q-CRAFT's production function after the WEO forecast horizon. Higher population growth means a larger labor force, higher potential GDP, and, all else equal, more favorable debt dynamics. Countries facing demographic decline see slower growth and more challenging fiscal trajectories. Total population growth also affects expenditure projections, since primary spending grows with total population (see Chapter 2 for the expenditure growth formula).

How to set it:

- **Medium** is the standard assumption for most policy analysis. Use it unless you have a specific reason to explore demographic scenarios.
- **High** and **Low** bracket uncertainty. They are useful for countries where fertility trends are uncertain or where demographic policy changes are anticipated.
- For countries experiencing rapid demographic transition (declining working-age population), compare Medium and Low to understand the fiscal sensitivity to demographic outcomes.

For detailed methodology, see the IMF User Guide, pp. 10-12 and Section IV.A on demography and employment.

Predict, observe, explain

Predict. Uganda has one of the youngest populations in the world. Before you touch the control: will switching from Medium to High move Uganda's 2099 debt ratio a lot, a little, or barely at all? Write down a direction and a rough size in percentage points.
Observe. Switch between Medium and High and watch the Baseline tab's debt trajectory. Then do the same for a country with a shrinking working-age population and compare the size of the move.
Explain. In countries with young, growing populations the difference between variants is small over the projection: the working-age share is already rising in every variant. In countries facing demographic decline the gap widens sharply, because slower working-age population growth means lower GDP growth and rising expenditure-to-GDP ratios at the same time. Note that both effects run through g, and that the expenditure channel runs through total population while the growth channel runs through working-age population. Those two populations move differently, which is why the net effect is not obvious in advance.

! Defend your choice: demography

A colleague from the Climate Finance Unit argues you should use the Low variant, because Uganda's fertility rate has been falling faster than the UN projected. Do you switch?

A defensible answer

Probably not for the headline run, and definitely worth a sensitivity run. The reasoning: Medium is the reference variant, and using anything else as your headline puts you in the position of defending a demographic forecast inside a fiscal risk document, which is not the argument you want to be having. The colleague's point is testable in two minutes. Run Low, record the difference in the 2099 debt ratio, and put the number in the packet as a sensitivity. If the difference is small, you have retired the objection with evidence. If it is large, you have found something worth a paragraph.

i Document it

Record: **variant chosen**, **why** (one line), and **the 2099 debt ratio under at least one alternative variant** as a sensitivity.

4.6.3 Debt target (% of GDP)

What it is. The debt-to-GDP ratio that the fiscal rule aims to converge toward over time. Range: 0-200% of GDP.

Why it matters. This parameter only has an effect when the fiscal rule is enabled (see below). It determines how aggressively government spending adjusts: a lower target relative to current debt means more consolidation is needed, which reduces primary expenditure. A higher target means less adjustment.

How to set it:

Start from the country's actual fiscal framework. Many countries have explicit fiscal rules with legislated debt targets: use those where they exist. Uganda's FY2024/25 Fiscal Risk Statement reads its climate results against a 50 percent of GDP fiscal rule ceiling, and reports that public debt passes it in the High, Hot and Vulnerable scenarios. Where no formal target exists, the debt-stabilizing primary balance, a standard DSA metric (see the IMF User Guide, Section IV.A, and Chapter 3), provides a useful benchmark.

As rough starting points (not authoritative IMF guidance):

- **Low-income countries (LICs):** 40-50% of GDP is a common range in practice, broadly consistent with IMF-World Bank Debt Sustainability Framework thresholds.
- **Emerging markets (EMs):** 50-70% of GDP, depending on the country's market access, institutional quality, and fiscal track record.
- **Advanced economies (AEs):** Can sustain higher ratios (60-100%+), though the appropriate target depends on growth prospects and market confidence.

For detailed methodology, see the IMF User Guide, pp. 15-18 on fiscal rule assumptions and the baseline scenario.

Predict, observe, explain

Predict. With the fiscal rule enabled, you are about to move the debt target from 40 percent to 80 percent. Write down what happens to primary expenditure in the early years, and what happens to the 2099 debt ratio.
Observe. Set the target to 40, read the primary expenditure path and the debt path in the Baseline tab. Set it to 80 and read them again.
Explain. A lower target forces faster consolidation: expenditure is cut more aggressively to bring debt down. A higher target leaves more spending room. The adjustment is gradual rather than immediate, because it works through the fiscal gap mechanism over multiple years. If you predicted an instant jump in expenditure, that is the thing to correct: the rule is a feedback loop keyed to last year's debt position, not a switch.

Defend your choice: the target

Your minister's office asks you to run the projection against a 40 percent target rather than the 50 percent ceiling the Fiscal Risk Statement uses, "to be prudent." What do you do, and what do you say?

A defensible answer

Run both, and be explicit about what the target is doing in the model. In Q-CRAFT the debt target is not a judgment about what is sustainable; it is the anchor the simulated fiscal rule steers toward. Changing it changes the assumed policy reaction, not the country's risk. The honest framing: "At the 50 percent anchor, the Hot scenario breaches the ceiling by X. At a 40 percent anchor, the consolidation required to stay inside it is Y percent of GDP a year of additional primary balance." That gives the office both numbers and does not smuggle a policy preference into a risk assessment.

Document it

Record: **target used**, **its source** (legislated rule, Charter for Fiscal Responsibility, DSF threshold, analyst judgment), and **the alternative target you tested**.

4.6.4 Fiscal rule (Yes / No)

What it is. When enabled, the fiscal rule adjusts primary expenditure to steer debt toward the target ratio. When disabled, spending follows baseline trends regardless of the debt level.

Why it matters. Without a fiscal rule, debt dynamics are purely mechanical: they depend on the interest-growth differential and the primary balance that results from revenue and expenditure trends. With the fiscal rule, the model simulates active fiscal policy: the government responds to rising debt by cutting spending or responds to fiscal space by allowing more spending.

How to set it:

- **Start with the fiscal rule off** to see the "no policy change" baseline. This reveals the underlying fiscal trajectory: is debt stable, rising, or falling under current trends?

- **Turn the fiscal rule on** to see how active fiscal policy changes the picture. This answers: "How much consolidation would be needed to reach the debt target?"
- Most countries in practice have some form of fiscal anchor, though enforcement varies. The IMF User Guide notes that Q-CRAFT's fiscal rule is a partial-equilibrium approximation: it does not model the GDP effects of fiscal consolidation (User Guide, p. 17, footnote 11).

The adjustment is applied after the multiplicative growth factors and is additive in levels: it modifies the level of primary expenditure, not the growth rate. The adjustment depends on the prior year's state, which is why Q-CRAFT computes this recursively year by year.

Predict, observe, explain

Predict. Run Uganda with the rule off first. Predict whether turning the rule on will change the 2099 debt ratio a lot or a little, and say why in terms of where the off-rule path ends up relative to the target.
Observe. Run the same country with the fiscal rule on and off. Watch both the debt path and the primary expenditure path.
Explain. With the rule off and unfavourable debt dynamics (interest rate exceeding growth), debt-to-GDP can rise without bound. Turn the rule on and debt converges toward your target, with the primary expenditure chart showing the spending cuts required to get there. The gap between the two runs is the fiscal effort implied by the target. If the off-rule path already sits near the target, turning the rule on does almost nothing, and that is itself a finding worth reporting.

! Defend your choice: rule on or off

Which run belongs in a Fiscal Risk Statement: rule on or rule off?

A defensible answer

Both, and they answer different questions. Rule **off** is the risk statement: this is where we end up if policy does not respond, which is the honest baseline for a document whose job is to disclose risk. Rule **on** is the policy statement: this is the effort required to hold the anchor. Publishing only the rule-on run understates the risk, because it assumes the response you are trying to justify. Publishing only the rule-off run invites the objection that no government would sit still for seventy years. Report the pair and say which is which.

Document it

Record: **rule on or off for the headline run**, **the paired run**, and **one sentence on which question each answers**.

4.6.5 Expenditure rigidity (0.0 - 1.0)

What it is. The degree to which government spending resists downward adjustment when climate shocks reduce GDP. This parameter only affects climate scenarios, not the baseline.

Why it matters. This is the key parameter for understanding climate-fiscal risk. When climate change reduces GDP, government revenue falls (since revenue tracks nominal GDP). What happens to spending determines how much of the climate cost shows up as additional debt.

- **1.0 (fully rigid, worst case):** Spending stays at its baseline level in local currency terms, even as GDP falls. Revenue declines but expenditure does not adjust. The entire climate impact passes through to the primary balance and onto the debt ratio. This represents a government that cannot or will not cut spending in response to slower growth, due to large civil service wage bills, entitlement programs, or political constraints on adjustment.
- **0.0 (fully flexible):** Spending adjusts proportionally with GDP decline, maintaining the same expenditure-to-GDP ratio as in the baseline. The debt impact of climate change is smaller because the government absorbs the shock by reducing real spending per capita.
- **Values between 0 and 1** represent partial adjustment. Most countries fall somewhere in between: some spending categories (wages, pensions, debt service) are rigid, while others (capital investment, discretionary programs) can adjust.

How to set it:

- Countries with large public sector wage bills, universal social programs, or rigid entitlement commitments: lean toward 0.7-1.0.
- Countries with more fiscal flexibility, smaller governments, or well-established expenditure review processes: 0.3-0.6.
- The default of 1.0 is deliberately conservative: it shows the worst-case scenario where all climate damage translates into additional borrowing.

For detailed methodology, see the IMF User Guide, pp. 20 and 35-36 on the expenditure rigidity parameter and fiscal effects of climate change.

> ⚠ WIDGET-TODO: expenditure rigidity
>
> Embed the expenditure rigidity intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The integration pass happens after this restructure and run 3 both land; this callout is the anchor point. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

> Predict, observe, explain
>
> **Predict.** On the Analysis tab you are about to compare the climate fan at rigidity 1.0 against rigidity 0.0. Write down which one produces the wider fan, and by roughly what factor.
> **Observe.** Set rigidity to 1.0 and read the 2099 gap between the baseline and Hot Unadapted. Set it to 0.0 and read the same gap.
> **Explain.** At 1.0 the fan spreads wide: the whole climate revenue loss accumulates as debt, so the gap between Paris-aligned and Hot Unadapted is large. At 0.0 the scenarios compress toward the baseline, because the government absorbs the shock through spending. The distance between those two fans is your country's fiscal exposure to expenditure rigidity, and it is usually the largest single lever in the tool. If your prediction was that rigidity matters less than the choice of climate scenario, check it: for most countries it does not.

⚠ SCREENSHOT-TODO

Side-by-side Analysis tab for Uganda at rigidity 1.0 and rigidity 0.0, with the same axis limits on both, so the compression is visible rather than described. Same-axis is the point; two auto-scaled charts will hide the effect.

❗ Defend your choice: rigidity

This is the judgment call you will actually be challenged on. A colleague says 1.0 is alarmist for Uganda: "no government holds spending flat in shillings for seventy years while the economy shrinks." Another says anything below 1.0 assumes a fiscal discipline Uganda has not demonstrated. What do you do?

DRAFT FOR TEAL: defending a rigidity choice

Both colleagues are making empirical claims, and the parameter is where you settle them with evidence rather than adjectives.

The evidence to reach for is the composition of the budget, not a view about political will. What share of primary expenditure is wages, pensions, statutory transfers and other commitments that require legislation to change? That share is close to a floor on rigidity. In most low-income countries it is high, and the flexible remainder is dominated by domestically financed development spending, which is exactly the spending a government protects last.

The defensible position is to run 1.0 as the headline and a lower value as the sensitivity, and to say why in one line: the default is conservative by design, and a fiscal risk statement is the document where conservatism belongs. Then give the second colleague their number: at rigidity 0.6, the 2099 gap narrows from X to Y percentage points.

What not to do is pick a middle value because it feels balanced. 0.5 with no reasoning behind it is weaker than 1.0 with a stated rationale, because 1.0 at least has the tool's own conservative-default argument attached. An unexplained middle number is the assumption that gets quietly reused for five years by people who never knew it was a guess.

The sentence for the packet: "Rigidity set at 1.0, the tool default, because roughly N percent of Uganda's primary expenditure is wages, pensions and statutory commitments; the sensitivity run at 0.6 changes the 2099 climate-baseline debt gap from X to Y percentage points."

ℹ **Document it**

Record: **rigidity used**, **the budget-composition evidence behind it**, and **the sensitivity run and its effect on the headline gap**. This is the entry the capstone rubric weights most heavily.

4.7 Wrapper: what you can now do

- You can set all five controls for a country and give a reason for each that cites evidence rather than preference.

- You can predict each parameter's direction of effect before moving it, which is what tells you when the tool is behaving oddly.
- You have five **Document it** entries. Together they are the assumption-rationale section of your export packet.

Common errors on this material: inverting the rigidity scale (see Chapter 1, question 2); moving the debt target while the fiscal rule is off and wondering why nothing happened; choosing a middle value to avoid an argument.

On your desk this week: find one number in a projection your department published and ask who chose it and why. If the answer takes more than a minute to find, you have just made the case for the export packet.

Chapter 5

Uganda end to end: from assumptions to the Fiscal Risk Statement paragraph

What you need to know:

- This module runs one country all the way through, from parameter choices to publishable prose
- The target format is not a chart. It is the write-up: the IMF's C-PIMA presentation of Q-CRAFT results, and Section III of Uganda's own Fiscal Risk Statement.
- The scaffolding fades: a full worked case, then a variant where you finish the interpretation, then one you do alone

Fast path

There is no fast path through this module. It is the module the capstone is built from. If you are short of time, do Section 5.6 and Section 5.8 and skip the middle.

5.1 Warm-up

From Chapter 4 and Chapter 2:

1. Which two parameters change the climate result but not the baseline?
2. In which year do climate scenarios begin to diverge from the baseline?

(Answers: expenditure rigidity, and the climate scenario itself. Everything else moves both. 2030.)

5.2 The hook: two documents that already exist

You are not inventing a format. Two published documents show what Q-CRAFT output looks like when it reaches a policymaker.

The IMF's C-PIMA write-up. *Uganda: PFM Climate Assessment: Public Investment and Fiscal Risks Management* (IMF, 2024) reports the September 2023 workshop with MoFPED staff in a single paragraph: the impact of climate change appears milder in Uganda than in other sub-Saharan African countries, but it matters for growth and fiscal sustainability, and by end of century GDP loss could pass 4 percentage points with debt at 66 percent of GDP against 47.5 percent in the baseline. One paragraph. Three numbers. A comparative judgment.

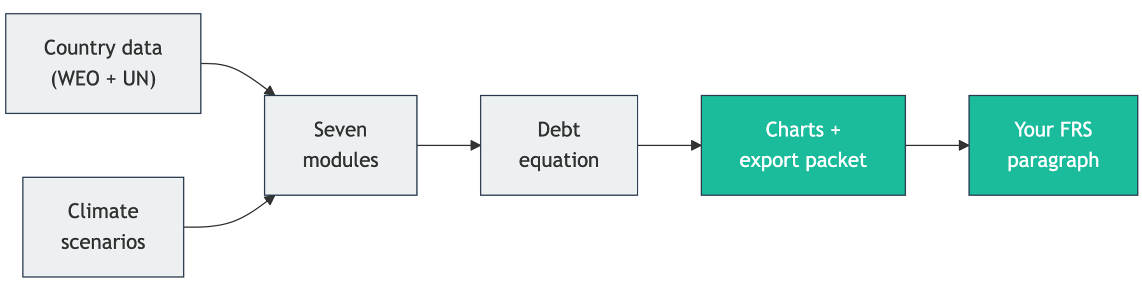
Uganda's Fiscal Risk Statement FY 2024/25. Section III, pages 13 to 17, is a fuller treatment: scenario definitions, a baseline fiscal path table, GDP deviation charts, fiscal balance and debt charts, and a closing policy sentence. The source line reads "QCRAFT (2023)."

Your capstone is the shorter of the two. This module builds it.

5.3 By the end of this module you can

- **Execute** a complete Q-CRAFT run for a country, from parameter selection to CSV export
- **Apply** the Baseline Sanity Check before interpreting any climate result
- **Interpret** the three chart families, saying what each one licenses you to claim
- **Draft** a two-paragraph fiscal risk write-up in the register of a published Fiscal Risk Statement
- **Assemble** the export packet: inputs, rationale, output

5.4 Where you are in the course



This module is the output node: charts, packet, and the prose that comes out of them.

5.5 The target format

Section III of Uganda's Fiscal Risk Statement FY 2024/25 is organised in three moves. Copy the structure; it is the one the ministry already publishes.

Move	What it contains	Where your material comes from
1. Set up the analysis	What the tool does, the scenarios used, and the baseline fiscal path	Chapter 4 parameters, Baseline tab
2. Macroeconomic effects	GDP deviation from baseline by scenario, and the productivity channel behind it	Climate tab

Move	What it contains	Where your material comes from
3. Fiscal effects	Primary and overall balance by scenario, debt-to-GDP by scenario, read against the fiscal anchor	Analysis tab

The published baseline table is worth copying too. Uganda's Table 5 reports four years, not seventy:

Summary fiscal path	2023	2050	2075	2099
Primary expenditure (% of GDP)	19.9	19.6	18.8	19.4
Interest expenditure (% of GDP)	3.3	1.3	1.8	2.4
Interest rate (%)	7.7	3.9	5.5	5.5
Primary balance (% of GDP)	-6.3	-1.7	-0.8	-1.4
Overall balance (% of GDP)	-9.6	-3.0	-2.7	-3.8
Debt-to-GDP ratio (%)	47.1	36.2	35.8	47.5

Source: Uganda Fiscal Risk Statement FY 2024/25, Table 5, sourced to QCRAFT (2023).

Four columns is a deliberate editorial choice: a seventy-row table is not a policy document. Note also what that baseline says before any climate damage is applied. Debt falls to the mid-30s by mid-century and comes back to 47.5 by 2099. Read the interest rate row against the growth path and you can see why.

5.6 The worked case: Uganda, start to finish

Follow along in the app. Every step names what you are doing and what it licenses you to say.

5.6.1 Step 1: set the parameters, and write down why

Open Q-CRAFT Explorer and select Uganda. Use these settings for the worked case:

Parameter	Setting	Rationale for the packet
Country	Uganda	Analysis subject
Demography variant	Medium	Reference variant; Low tested as sensitivity
Debt target	50% of GDP	The ceiling Uganda's own Fiscal Risk Statement reads results against
Fiscal rule	Off for the headline run	Discloses the risk without assuming the policy response; rule-on run reported alongside

Parameter	Setting	Rationale for the packet
Expenditure rigidity	1.0	Tool default, conservative; budget composition evidence per Chapter 4; 0.6 tested as sensitivity

That table is the first page of your export packet. Write it before you look at any output, so the settings are choices rather than post-hoc justifications.

5.6.2 Step 2: read the Baseline tab

The Baseline tab shows the no-climate-change scenario: the country's fiscal trajectory under current trends and the assumptions you have set.

Debt-to-GDP trajectory (top). The headline chart. Follow the trajectory from the historical period (shaded, through 2029) into the projection period. Is debt rising, stable, or falling? If the fiscal rule is on, debt should converge toward your target. If it is off, the trajectory reveals the underlying fiscal dynamics: favourable (growth exceeds interest rates) or unfavourable (interest rates exceed growth).

Revenue and Expenditure (% of GDP) (bottom left). Revenue-to-GDP stays constant by assumption: it grows with nominal GDP. Primary expenditure-to-GDP may diverge because expenditure grows with total population and productivity, while revenue grows with working-age population. In countries where working-age population grows faster than total population (demographic dividend), expenditure-to-GDP falls and fiscal space opens. In ageing countries, expenditure-to-GDP rises, a structural fiscal pressure independent of any policy choice.

Fiscal Balances (bottom right). The primary balance (revenue minus non-interest expenditure) and overall balance (including interest payments). The zero line is the key reference: a primary surplus means the government is generating enough non-interest revenue to cover non-interest spending. The gap between primary and overall balance reflects the interest burden on existing debt.

> ⚠ SCREENSHOT-TODO
>
> Baseline tab for Uganda, all three charts, with callouts on: (a) the end of the shaded WEO period at 2029, (b) the 2099 debt ratio, (c) the primary-to-overall balance gap. This is the image the whole module refers back to.

5.6.3 Step 3: run the sanity check before you interpret anything

> Baseline Sanity Check
>
> Before interpreting climate results, verify the baseline makes sense:
> - ☐ Does the initial debt level match your country's actual current debt-to-GDP ratio?
> - ☐ Is the revenue-to-GDP ratio plausible given WEO forecasts?
> - ☐ Does the expenditure path reflect realistic spending trends?

☐ If the fiscal rule is on, does debt converge toward your target?
☐ Do the primary and overall balance paths make economic sense given the country's fiscal history?

! DRAFT FOR TEAL: the sanity check applied to Uganda

Run the five boxes against what is on your screen and against the published figures.
Initial debt level. The published baseline puts 2023 at 47.1 percent of GDP. Uganda's Fiscal Risk Statement reports 46.9 percent at June 2023 on the ministry's own basis. Those agree. If your Explorer run shows something materially different, the likely cause is the WEO vintage: the Explorer ships October 2024, the published table came from the 2023 run. Note the difference, do not fix it by overriding data.
Revenue-to-GDP. Revenue is held at a constant share of GDP by construction. The check is not whether it moves; it is whether the level it is held at is the right one. For a country in the middle of a revenue mobilisation strategy, a flat revenue ratio is a conservative assumption and is worth one sentence in the write-up.
Expenditure path. The published table shows primary expenditure drifting from 19.9 to 19.4 percent of GDP over seventy-six years, dipping to 18.8 around 2075. That is the demographic dividend showing up: total population growth drives spending, working-age population growth drives GDP, and for a while the second outruns the first. If your expenditure path rises instead of drifting down, check the demography variant.
Fiscal rule convergence. With the rule off, skip this box, and say in your packet that you skipped it and why.
Balance paths. The published baseline has the primary deficit narrowing from 6.3 percent of GDP in 2023 to 1.4 percent by 2099, with the overall deficit at 3.8. That is a large near-term consolidation built into the WEO forecast period, not something the model invented. Whether it is credible is a question about the WEO, and it belongs in your write-up as a caveat rather than a correction.
The rule this step encodes: a climate result computed on a baseline you have not checked is a number with no owner. Every one of the five boxes is a question someone will ask you.

5.6.4 Step 4: read the Climate tab

The Climate tab shows how climate change affects real GDP under six warming scenarios, using empirical estimates from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on Kahn et al. (2021).

What to look for. Two charts show absolute GDP levels and a GDP index (2029 = 100). The six scenarios span from Paris-aligned (below 2°C warming by 2100) to Hot Unadapted (high warming with slow adaptation). Climate impacts begin in 2030; before that year, all scenarios match the baseline. The gap between scenarios represents the range of GDP outcomes across different climate futures.

Countries closer to the equator generally show larger GDP losses. The empirical estimates reflect historical relationships between temperature deviations and economic output: countries already in warm climates experience more severe productivity impacts from additional warming.

A country showing a large gap between Paris and Hot Unadapted scenarios faces high climate-fiscal vulnerability. A small gap suggests lower direct climate exposure in the model, though this does not account for indirect effects (natural disasters, sea-level rise, trade disruptions) that Q-CRAFT's conservative methodology does not capture (User Guide, p. 5).

! DRAFT FOR TEAL: what the Uganda GDP chart licenses you to say

The published result is a level GDP loss of around 4 percent by end of century under the Hot scenario, and the C-PIMA summary describes Uganda's impact as milder than other sub-Saharan African countries.
Four percent is a number that needs a perspective sentence, because on its own it reads as small. Two are available and they point in opposite directions, so pick deliberately and say which you picked.
The deflating one: 4 percent of GDP spread over seventy-five years is a few hundredths of a percentage point off annual growth, well inside the noise of any single year's outturn.
The alarming one: it is a permanent level loss that compounds into the debt ratio every year thereafter, and 4 percent of GDP is a recurring annual amount comparable to a large sector budget.

> **TODO: verify the perspective figure**
> The second framing needs a checked comparator before it ships: 4 percent of Uganda's GDP expressed against a named line of the approved budget (health, or the road sector), sourced to the Approved Estimates for the relevant year. Do not ship an approximate comparison in a document aimed at the ministry that publishes the actual number.

Both are true. The second is the one a fiscal risk statement is for, and the first is the one you should expect back from a sceptical reader. Write the paragraph so it survives both.
What the chart does not license: any statement about what Uganda's GDP will be. This is a deviation between two runs of the same model, not a growth forecast.

⚠ SCREENSHOT-TODO

Climate tab for Uganda, GDP index chart, with the six scenario lines labelled on the data rather than in a legend, and the 2030 divergence point marked.

5.6.5 Step 5: read the Analysis tab

The Analysis tab is the comparison view: it overlays baseline and all climate scenario debt trajectories on a single chart.

The climate-fiscal risk premium. The spread between the baseline debt trajectory and the climate scenario trajectories is the country's climate-fiscal risk premium: the additional debt burden attributable to climate change. This spread depends on three factors:

1. **Climate exposure:** How much GDP the country loses under each scenario (from the Climate tab).
2. **Expenditure rigidity:** How much of the GDP loss passes through to the primary balance and debt. Higher rigidity means a wider spread.
3. **Starting fiscal position:** Countries with unfavourable baseline debt dynamics (interest exceeding growth) see climate shocks amplified through the debt accumulation equation.

Reading the chart. If the baseline shows stable or declining debt but the Hot Unadapted scenario shows rapidly rising debt, climate change has the potential to destabilize an otherwise sustainable fiscal position. This is the core finding Q-CRAFT is designed to surface: it answers the question, "Even if our fiscal policy is sound today, could climate change make it unsustainable?"

> **!** DRAFT FOR TEAL: reading the Uganda fan chart
>
> Three readings, in the order a reviewer will want them.
> **The gap.** Baseline at 47.5 percent of GDP in 2099, Hot at 66. Eighteen and a half percentage points is the headline, and it is the number that goes in the first sentence. Uganda's own Fiscal Risk Statement says the same thing from the other end: the primary deficit in the Hot scenario ends 0.7 percentage points worse than baseline, "thereby raising public debt by over 18 percent of GDP." A 0.7 point annual deficit gap producing an 18 point debt gap is the compounding from Chapter 3 doing its work, and it is worth saying out loud, because it is the part non-specialists miss.
> **The threshold.** The gap only becomes a policy fact when it crosses something. Uganda's Statement reports that in the High, Hot and Vulnerable scenarios debt passes the 50 percent of GDP fiscal rule ceiling. That sentence is doing more work than the 18 points: it converts a projection into a breach of an existing commitment.
> **The shape.** Look at when the lines separate, not just where they end. Scenarios are identical until 2030, then fan slowly. A path that crosses the ceiling in 2085 and one that crosses in 2060 imply very different urgency, and the fan chart shows the difference at a glance where a 2099 table does not.
> **What to do when the baseline itself is rising.** If your run has debt rising in the baseline, the climate gap is no longer the interesting number, because both paths are unsustainable. Say so, and report the year each path crosses the anchor instead.

5.6.6 Step 6: export the packet

The Data tab provides an interactive data grid and CSV export functionality. Two download options are available:

- **Download Baseline CSV:** The baseline fiscal projection (no climate scenarios). Useful for incorporating Q-CRAFT projections into your own fiscal framework or DSA.
- **Download All Scenarios CSV:** Baseline plus all six climate scenarios, stacked with a scenario identifier column. Use this for custom analysis, cross-country comparison, or integration with other tools.

All values are in billions of local currency units (except ratios, which are percentages of GDP). The data covers the full projection period from 2009 through 2099.

Download both. Add them to the parameter table from Step 1 and the Document it entries from Chapter 4. That is the export packet: inputs, rationale, output, in one place, reproducible by someone who has never met you.

5.6.7 Step 7: write the two paragraphs

❗ DRAFT FOR TEAL: model two-paragraph write-up

Built from the published Uganda figures so every number in it is checkable. Learners replace them with their own run.

> Climate change presents a measurable long-term risk to Uganda's fiscal position. Using the IMF's Quantitative Climate Risk Assessment Fiscal Tool, the baseline projection, which assumes no climate damage, has public debt at 47.5 percent of GDP in 2099 after falling to around 36 percent by mid-century as the demographic dividend widens the gap between growth and the effective interest rate. Under the Hot scenario, in which emissions follow SSP3-7.0 and temperatures track the 90th percentile of model projections, output is around 4 percent below baseline by the end of the century and public debt reaches 66 percent of GDP. The mechanism is indirect: higher temperatures slow labour productivity growth, slower growth reduces revenue, and with primary expenditure held at its baseline level in shilling terms the shortfall accumulates as debt. An annual primary deficit only 0.7 percentage points wider than baseline compounds into a debt ratio more than 18 percentage points higher over seventy years.
>
> The policy significance lies in the threshold rather than the magnitude. In the High, Hot and Vulnerable scenarios, public debt passes the 50 percent of GDP fiscal rule ceiling and takes on a rising trajectory, which means climate damage alone is sufficient to move Uganda from compliance to breach without any other adverse shock. These estimates are conservative. They capture the slow productivity effects of temperature change and exclude natural disasters, sea-level rise, non-market damages and the public expenditure required to adapt, so the true fiscal exposure is larger than reported here. The results are also sensitive to the expenditure rigidity assumption: at full rigidity, the tool's default, the entire revenue loss becomes borrowing, while at 0.6 the end-century gap narrows materially. Meeting Uganda's Paris commitments and building expenditure flexibility both reduce this exposure, and the two are complements rather than alternatives.

What makes this a Fiscal Risk Statement paragraph rather than a report of a model run:

1. It names the tool and the scenario in one clause each, then moves on.
2. Every number is a comparison between two runs, never a forecast.
3. It gives the mechanism in one sentence, so a reader who does not know the debt equation can still follow the causal chain.
4. It states the threshold, because a breach of an existing commitment is what makes a projection actionable.
5. It states the conservatism caveat before anyone else can, which is what makes the rest credible.
6. It ends on something the ministry can do.

What Teal should rewrite: the second paragraph's closing sentence is doing policy advocacy in a risk document, and the right register for that is a judgment call I should not be making.

5.7 Completion problem: the High scenario variant

Same country, same parameters, one change: report against the **High** scenario rather than Hot. The first steps are done for you. Finish the interpretation.

Done for you:

- Parameters as in Step 1, unchanged
- Baseline unchanged, so the sanity check from Step 3 still holds and does not need repeating
- High is SSP3-7.0 at the average of model temperature projections, rather than the 90th percentile

Your turn:

1. Read the 2099 debt ratio under High from the Analysis tab. Record it. ______
2. Compute the gap to baseline. ______ percentage points.
3. Read the year the High path crosses 50 percent of GDP, if it does. ______
4. Write one sentence explaining why High sits below Hot when both use the same emissions pathway.
5. Rewrite the first sentence of the model paragraph above using your High numbers.

> **i** Checking yourself on step 4
>
> The distinction is not emissions, it is the temperature distribution. High takes the average temperature outcome across climate models running SSP3-7.0; Hot takes the 90th percentile of that same set. Same policy world, different draw from the uncertainty. Which one belongs in a fiscal risk statement is a real question: risk documents conventionally report adverse tail outcomes, which argues for Hot, while a central-case fiscal framework argues for High. Report both and say why you led with the one you led with.

5.8 Independent problem: the adaptation-speed variant

No scaffolding this time. Checklist only.

The task. Compare **Hot + Adapted** against **Hot + Unadapted** for Uganda. Same temperatures in both; the difference is how quickly the economy adjusts.

Deliver:

- ☐ The 2099 debt ratio under each, and the gap between them
- ☐ One sentence on what that gap represents in policy terms
- ☐ One sentence on what it does **not** represent, which is the trap in this comparison
- ☐ The same comparison at expenditure rigidity 0.6, with a note on whether the adaptation gap or the rigidity gap is larger for Uganda
- ☐ A two-paragraph write-up in the format of Step 7

> **i** The trap, if you want to check before you write
>
> Faster adaptation in Q-CRAFT is a faster economic adjustment to a given temperature path. It is not adaptation spending. The model does not cost the investment that would produce

that adjustment, and the User Guide says so directly (p. 6). So the Adapted-to-Unadapted gap is the benefit side of an adaptation programme with the cost side missing. Reporting it as the value of adaptation without that sentence attached is the single most likely way to misuse this tool in a budget discussion.

5.9 Wrapper: what you can now do

- You have run a country end to end and produced an export packet with rationale attached.
- You can apply the Baseline Sanity Check and say what each box protects you from.
- You can write two paragraphs that a Fiscal Risk Statement editor would recognise.

Common errors on this material: interpreting climate results before checking the baseline; quoting a 2099 level as a forecast; reporting the adaptation gap as the value of adaptation.

On your desk this week: take the two paragraphs you drafted to whoever owns your ministry's fiscal risk chapter and ask what they would cut. The answer tells you more about the format than this module can.

Chapter 6

Know what the tool cannot tell you

What you need to know:

- Q-CRAFT's estimates are conservative by construction, and the list of what they exclude is specific enough to put in a footnote
- The baseline floors debt at zero and the climate scenarios do not, which changes how you read the bottom of a fan chart
- Two countries with the same debt path can have completely different underlying drivers, and the tool will not tell you which one you are looking at
- Q-CRAFT and the LIC-DSF answer different questions. Knowing which one you need is a skill, and it runs in both directions.

Fast path

No fast path. This is the shortest module and it is the one that keeps you out of trouble. If you only read one section, read Section 6.6.

6.1 Warm-up

From Chapter 5:

1. What does the gap between Hot Adapted and Hot Unadapted represent, and what does it leave out?
2. Which check comes before any climate interpretation?

(Answers: the benefit of faster adjustment, with the cost of achieving it not modelled. The Baseline Sanity Check.)

6.2 The hook: the question after the presentation

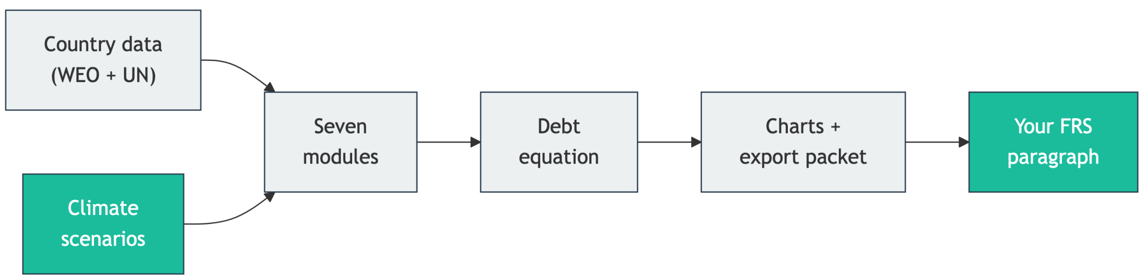
You have presented the fan chart. The Permanent Secretary looks at the Hot Unadapted line and asks: "So this is what climate change will cost us?"

The whole of your credibility rests on the next thirty seconds. This module is that answer, in its several parts.

6.3 By the end of this module you can

- **List** the specific damage channels Q-CRAFT excludes, and **state** the direction of the resulting bias
- **Explain** why baseline and climate scenarios treat negative debt differently, and **adjust** your reading of a chart accordingly
- **Compare** two countries with similar debt paths and **distinguish** the different drivers behind them
- **Decide** whether a given question calls for Q-CRAFT, the LIC-DSF, both, or neither, and **justify** the choice

6.4 Where you are in the course



This module is about the scenarios' limits and what they license in the write-up.

6.5 What the numbers mean, and what they do not

Q-CRAFT Explorer produces stylized long-term projections. The results are useful for understanding the direction and magnitude of climate-fiscal risks, comparing scenarios, and identifying the fiscal parameters that matter most for a given country.

The results are not forecasts. Projections to 2099 depend on assumptions about productivity, inflation, interest rates, and climate that involve deep uncertainty. The value is in the comparison across scenarios and the sensitivity to assumptions, not in any single projected number.

Q-CRAFT is designed to complement, not replace, existing fiscal analysis. The IMF User Guide positions it as "a starting point for climate change fiscal risk analysis" that supplements country macroeconomic models (User Guide, pp. 5-6). Use Q-CRAFT to identify which climate-fiscal channels matter most for your country, then investigate those channels with more detailed models and country-specific data.

6.6 The conservatism list

The scenarios capture one channel: the slow effect of temperature change on productivity, and through it on growth. Everything below is excluded, and every exclusion runs the same direction.

Excluded	Why it matters for a fiscal risk statement
Climate-induced natural disasters	The channel Uganda's own risk statement leads with: floods and epidemics were about 75 percent of recorded events 1985 to 2021
Sea-level rise	Not a Uganda channel; decisive for coastal and small-island economies
Tipping points	The estimates are fitted to historical temperature variation and cannot extrapolate to discontinuities
Non-market damages	Morbidity, mortality, conflict, food insecurity: real costs, absent from GDP
Spillover effects	Trade partners, remittance sources and commodity markets are hit too
Adaptation spending	Even the Hot Adapted scenario models faster adjustment, not the public expenditure that buys it (User Guide, p. 6)

The direction is what makes this list usable. Every exclusion is a cost left out, so the modelled fiscal impact is a lower bound on the modelled channels, not a central estimate of total impact. The sentence that follows from it: *actual fiscal impacts are likely larger than these results suggest.*

Say it yourself, in the write-up, before a reviewer says it for you. A conservative number with the conservatism disclosed is stronger than a larger number without it.

! DRAFT FOR TEAL: what conservatism does not buy you

Conservatism protects the credibility of a number. It does not make the number safe to quote out of context, and it creates one specific hazard worth naming.
A conservative estimate that gets picked up as *the* estimate becomes a ceiling in someone else's argument. "The analysis shows climate costs Uganda 18 points of debt" is a sentence that starts life as your careful lower bound and ends life as a budget office's reason to fund adaptation at 18 points' worth. The defence is structural rather than rhetorical: put the exclusion list in the same paragraph as the headline number, not in a footnote, and not in an annex.

6.7 The debt floor asymmetry

i The rule

The baseline scenario applies a floor of zero to the debt-to-GDP ratio: if the equation produces a negative value, debt is set to zero. Climate scenarios do not apply this floor, allowing debt to go negative (representing net asset positions). This asymmetry is intentional; it avoids masking the full range of climate-scenario outcomes.

DRAFT FOR TEAL: how the asymmetry changes what you are looking at

It matters for a narrow set of countries, and for those countries it matters a lot.
When it bites. Only when a projection drives debt to zero. That happens for countries with strongly favourable interest-growth differentials and sustained primary surpluses over a seventy-year horizon. It is not a Uganda problem on the published baseline, which lands at 47.5 percent.
What it does to the picture. In such a country the baseline flattens along zero while the climate scenarios continue downward into negative territory. Read naively, the chart shows climate scenarios with *lower* debt than baseline, which is nonsense. What you are actually seeing is one line clipped and the others not.
The reading rule. If any line in your fan chart touches zero, stop using the vertical gap as your headline number. Check the underlying series in the Data tab, and either report the comparison over a window before the floor binds or report the primary balance gap instead, which is not clipped.
Why the asymmetry exists rather than being a bug. Flooring the climate scenarios too would compress exactly the range the tool exists to show. The design choice preserves the spread and pushes the interpretive burden onto you. This module is where that burden gets paid.

6.8 Two countries, same chart, different problem

Transfer is the thing this course is actually for: recognising which situation you are in when the labels come off. Structured comparison is how it gets taught.

Comparison exercise: same path, different drivers

Find the pair. In the Explorer, work through countries in the dropdown until you find two whose baseline 2099 debt ratios are within about 5 percentage points of each other, but whose baseline-to-Hot-Unadapted gaps differ by 10 percentage points or more.
Then diagnose. For each country, decide which of the three drivers from Chapter 5 explains the difference:

1. Climate exposure (how much GDP is lost, from the Climate tab)
2. Expenditure rigidity (how much of that loss becomes debt)
3. Starting fiscal position (whether the interest-growth differential amplifies or damps it)

Write one sentence per country saying which driver dominates, and one sentence saying what policy conclusion follows for each. The conclusions should differ. If they do not, you have picked a pair that is too similar.
Then reverse it. Find two countries with similar climate exposure on the Climate tab whose debt outcomes diverge. The driver is now somewhere in the fiscal position, and finding it is the exercise.

TODO: seed the comparison with verified pairs

This exercise currently sends learners hunting. Two or three verified country pairs, run and checked against the current data vintage, would make it usable in a 20-minute workshop breakout. Needs a real run rather than an assumption about which pairs work.

DRAFT FOR TEAL: why this exercise and not a worked comparison

The instinct is to hand over the pair and the answer. Doing the search is the exercise, because the skill being built is diagnosis from surface features that mislead, and a country pair you were handed has already had the diagnosis done for you.

The general principle is the one to carry out of this module: a debt path is an outcome, not a diagnosis. Two countries arriving at 60 percent of GDP in 2099, one because it is highly exposed to warming and one because it borrows at rates above its growth, need different policies and different chapters in different documents. Q-CRAFT will show you the outcome. Deciding which country you are looking at is your job, and it is the part that cannot be automated.

6.9 Which tool applies here?

Q-CRAFT and the IMF-World Bank Low-Income Country Debt Sustainability Framework both produce debt projections for low-income countries. They are not substitutes, and using one where the other belongs is a common and expensive error.

	Q-CRAFT	LIC-DSF
Question it answers	How much does climate damage change the long-run fiscal path?	Is this country's debt sustainable, and at what risk rating?
Horizon	To 2099	Medium term, with longer-run tails
Output	Scenario comparison, no rating	A risk rating that carries operational consequences
Climate	Central: six scenarios	Not the organising principle
Debt detail	One aggregate stock	External and public splits, currency, creditor composition
Used for	Fiscal risk disclosure, C-PIMA annexes, long-term planning	Lending decisions, concessionality terms, programme design

Self-check: five questions, which tool?

For each, name Q-CRAFT, LIC-DSF, both, or neither, and say why in one line.

1. "The Fiscal Risk Statement needs a section on climate. What do we run?"
2. "Are we still at moderate risk of debt distress?"
3. "If we take this Eurobond instead of an IDA credit, what happens to our risk rating?"
4. "What is the fiscal cost of the National Adaptation Plan?"
5. "Does climate risk change our debt sustainability assessment?"

DRAFT FOR TEAL: self-check answers

1. **Q-CRAFT.** This is its designed use, and it is the use Uganda already made

of it in the FY2024/25 statement.

2. **LIC-DSF.** Q-CRAFT produces no risk rating and has no thresholds to breach. Answering this question from a Q-CRAFT chart is the error this section exists to prevent.
3. **LIC-DSF.** The question is about creditor composition and terms, which Q-CRAFT does not represent: it carries one aggregate debt stock and one effective interest rate.
4. **Neither.** Q-CRAFT does not cost adaptation, by design (User Guide, p. 6), and the LIC-DSF does not either. This question needs a costed adaptation plan, and the honest answer is to say so rather than to produce a number from a tool that cannot see it.
5. **Both, in sequence, and this is the interesting one.** Q-CRAFT establishes whether the climate channel is large enough to matter for the fiscal path. If it is, the finding belongs as a qualitative risk consideration alongside the DSF assessment rather than as an input to it, because the two are on different horizons and different definitions of debt. Anyone who tells you they have mechanically fed Q-CRAFT output into a DSF has done something that needs explaining.

The reverse direction, which is the part usually skipped: a question that arrives framed for the DSF may actually be a Q-CRAFT question. "Our risk rating is fine, so climate is not a debt issue" is a DSF answer to a long-horizon question the DSF does not ask. Uganda's own result makes the point: a baseline that is comfortable on any medium-term view still crosses the fiscal rule ceiling by end of century once climate damage is applied.

6.10 Wrapper: what you can now do

- You can list what the scenarios exclude and state the direction of the bias without looking it up.
- You can spot a clipped baseline in a fan chart and say why the vertical gap has stopped meaning what it usually means.
- You can take a question and say which tool it belongs to, in both directions.

Common errors on this material: quoting a conservative estimate as a total; reading a floored baseline as a favourable climate result; answering a sustainability-rating question from a Q-CRAFT chart.

The thirty-second answer to the Permanent Secretary, for reference: "No. It is what climate change does to the debt ratio through one channel, growth, under stated assumptions about how spending responds. It excludes disasters, sea-level rise and the cost of adapting, so it is a floor rather than a total. What it tells us is that the channel is large enough to breach the ceiling on its own."

On your desk this week: find a projection in circulation in your ministry and write down one question it cannot answer. Circulate that sentence with it.

Chapter 7

The capstone: your analysis, defended

What you need to know:

- The capstone is the export packet plus a two-paragraph Fiscal Risk Statement draft, for a country you choose
- It is marked on whether you can defend your assumptions, not on whether your numbers match anyone else's
- This module also closes the loop on Chapter 1, so you can see what changed

7.1 Warm-up

From across the course, without looking:

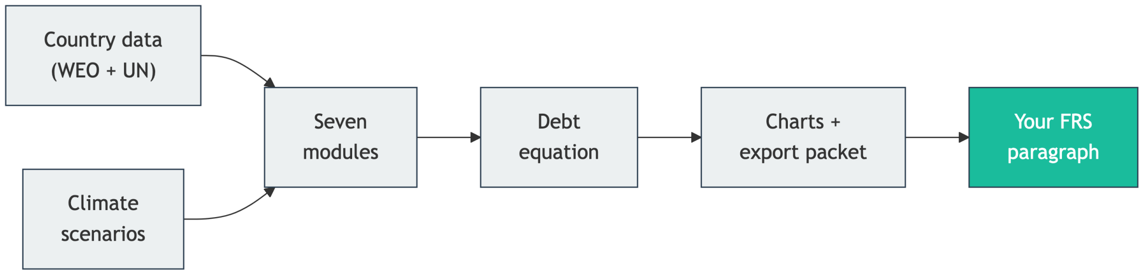
1. Name the two terms of the debt equation climate damage moves.
2. What does expenditure rigidity 1.0 mean?
3. Name one thing the climate scenarios exclude.

(Answers: g and pb. Spending is sticky, so the revenue loss becomes borrowing. Natural disasters, sea-level rise, tipping points, non-market damages, spillovers, or adaptation spending.)

7.2 By the end of this module you can

- **Produce** a complete export packet for a country of your choosing
- **Defend** each parameter choice against a specific challenge
- **Assess** your own draft against the rubric before anyone else sees it
- **Compare** your current capability against your Chapter 1 self-assessment

7.3 Where you are in the course



The last node: the document you hand over.

7.4 The capstone brief

The scenario. Your Permanent Secretary has asked for a climate-fiscal risk section for the next Fiscal Risk Statement. You have the Explorer, the WEO vintage it ships with, and whatever you know about the country. You have one week and half a page.

What to hand in.

1. **The export packet.**
   - The parameter table: every setting, with a one-line rationale each (the five **Document it** entries from Chapter 4)
   - The data vintage, stated
   - Both CSV exports, baseline and all scenarios
   - At least two sensitivity runs, with what changed and by how much
2. **The two-paragraph draft**, in the format of Chapter 5, Step 7.
3. **Three challenge answers.** Written responses of two to three sentences each to:
   - "Is this a forecast?"
   - "Why did you choose that rigidity value?"
   - "So this is what climate change will cost us?"

Country choice. Uganda if you want to compare against the published Section III. Any other country in the dropdown if you want a harder problem, and a country whose baseline behaves differently from Uganda's is a harder problem in a useful way.

What is not required. Matching the published Uganda figures. You are on a later WEO vintage and possibly different parameters, and reproducing someone else's numbers is not the skill being assessed.

7.5 The rubric

❗ DRAFT FOR TEAL: capstone rubric

Four criteria. The weights are a proposal, not a decision.

Criterion	Weight	Meets the standard	Falls short
Assumptions defended	40%	Every parameter has a rationale citing country evidence. Rigidity in particular has budget-composition reasoning behind it, and a sensitivity run.	Parameters listed without reasons, or reasons that restate the setting ("rigidity 1.0 because spending is rigid").
Baseline checked	20%	The five sanity-check boxes are answered against the actual run, including any that were skipped and why. Differences from the ministry's own figures are explained rather than reconciled away.	Climate results interpreted with no evidence the baseline was examined.
Interpretation	25%	Numbers presented as comparisons between runs. The mechanism stated in one sentence. A threshold identified. The conservatism caveat in the same paragraph as the headline number.	A 2099 level quoted as a prediction, a debt gap described as a cost, or the caveat relegated to a footnote.
Written for the reader	15%	Reads like the ministry's own risk chapter. A non-specialist can follow the causal chain. Two paragraphs, not five.	Reads like a description of a model run. Jargon undefined. Over length.

The single disqualifying error, worth naming separately because it recurs: reporting the Adapted-to-Unadapted gap as the value of adaptation. It is the benefit side with the cost side missing (Chapter 5, independent problem).

Note for the live workshop. The rubric doubles as the peer-review instrument. Three-pass protocol: first pass each reviewer marks assumptions only, second pass interpretation only, third pass the writing. Splitting the passes stops reviewers spending all their attention on prose, which is what happens when they are asked to review everything at once.

7.6 Common errors, harvested

> ⚠ TODO: replace with pilot-observed errors
>
> The list below is built from the misconceptions this course was designed against, not from watching people make them. The redesign plan treats a pilot with two or three target-profile testers as non-negotiable, and the error log from that pilot should replace this list. Until then, treat it as a hypothesis.

- Reading rigidity backwards, so the headline risk number comes out inverted
- Interpreting climate results before checking the baseline
- Quoting a 2099 level as a forecast
- Adjusting the debt target while the fiscal rule is off, then concluding the target does nothing
- Describing a debt-ratio gap as a cost in shillings
- Reporting the adaptation gap as the value of adaptation
- Answering a debt-sustainability-rating question from a Q-CRAFT chart

7.7 Where you started, where you are

Go back to your three answers from Chapter 1. Answer them again now, before reading on.

On long-term fiscal projection: A1 (not worked with a DSA) / A2 (can follow one) / A3 (could build one this week)

On climate-fiscal analysis: B1 (climate side only) / B2 (can describe the use) / B3 (have run or interpreted one)

On the tool itself: C1 (never opened it) / C2 (clicked around) / C3 (ran it and explained the output)

If you have completed the capstone, C3 is now true by construction: you have run it and written the explanation. The A and B rows are the ones worth looking at. Movement on those is the course working; no movement on those with C3 achieved means you can drive the tool and the economics underneath it did not land, which is worth a conversation rather than a shrug.

> Retake the three questions from Module 0
>
> Same three items, unchanged. Answer from memory before opening anything.
>
> 1. Zero primary balance, $r = 9$ percent, $g = 6$ percent. Debt ratio next year: falls, flat, or rises?
> 2. Expenditure rigidity 1.0 means what about spending under a climate scenario?
> 3. Debt at 66 percent under Hot against 47.5 percent baseline. What is the defensible statement of what that means?
>
>   **The answers, and what a wrong one now means**
>
>   1. **Rises**, to about 51.4 from 50. If this one is still wrong, Chapter 3 did not do its job and nothing in your capstone's interpretation section can be trusted, because every climate result in the tool is this arithmetic with a lower g.
>   2. **Spending stays at its baseline level in local currency terms**, so the whole revenue loss becomes borrowing. Getting this wrong at the end of the

course means your headline number may have the wrong sign on its sensitivity, which is the error most likely to reach print.
3. **Under stated assumptions, climate damage of this magnitude adds roughly 19 percentage points of GDP to the debt ratio by 2099 relative to the same projection without it.** A comparison between two runs, with the assumptions attached. If you reached for the forecast framing or the cost framing, reread Chapter 6 before you circulate anything.

7.8 What to use on your desk this week

- The debt-stabilizing primary balance for your country: three numbers, one division, and it opens most credible fiscal risk write-ups.
- The **Document it** habit: every assumption gets a line saying why. It costs a minute per parameter and it is the difference between analysis and a spreadsheet nobody can defend a year later.
- The which-tool question. When a request arrives, ask what question it is really asking before opening anything.

7.9 Spaced follow-up

Two things arrive after the workshop, and they are part of the course rather than an afterthought.

- **Day 3:** one retrieval question by email. No preparation, no marking.
- **Week 2:** an application check. What did you run, what did you find, what got used.

Spacing beats massing, and the IMF's own evaluation of capacity development found that training with hands-on follow-up sticks while the standalone workshop decays. The follow-up is the part that makes the rest count.

⚠ TODO: workshop materials

The live sessions need artefacts this guide does not yet contain: breakout task cards, the anonymous poll items, and the peer-review sheet derived from the rubric above. Tracked separately from the book.

7.10 Wrapper: the whole course in six lines

- One equation decides the debt path. Seven modules feed it. (Chapter 2)
- Two things move the ratio: the interest-growth gap, and the primary balance. (Chapter 3)
- Every parameter is a judgment call, and the record of the call is the deliverable. (Chapter 4)
- The output is a comparison between two runs, written up in the ministry's own format. (Chapter 5)
- The estimate is a floor, the baseline is floored at zero and the scenarios are not, and some questions belong to a different tool. (Chapter 6)
- What you hand over is an analysis you can defend. (this module)

Appendix A

Appendix: from Q-CRAFT to the LIC-DSF

This appendix is not part of the course

The modules teach you to run the tool and defend the analysis. This appendix is about the project behind the tool: why it was built, what it is a proof of concept for, and how to shape the next version. Nothing in the capstone depends on it. Read it if you want to influence what V2 does.

What you need to know:

- Q-CRAFT is a proof of concept for a larger goal: making the LIC-DSF more usable through human-centered design
- The biggest barrier to effective fiscal projection tools is ergonomics, not economics
- We are looking for co-design partners, not endorsements

A.1 The real barrier: ergonomics, not economics

The biggest barrier to effective use of fiscal projection tools is not the economics. It is the ergonomics. Smart people struggle not because the methodology is too complex, but because the tools do not guide them through the decisions they need to make.

This observation has emerged from years of conversations with MoF economists, central bank staff, and IFI economists. It has inspired a more organized human-centered design approach: structured design sprints and organized discussions with practitioners who use these tools in their daily work.

A.2 What we built and what we proved

Q-CRAFT Explorer reimplements the IMF's Q-CRAFT Excel workbook as an open-source Python web application. The calculation engine is a standalone package with automated parity tests against the original Excel tool.

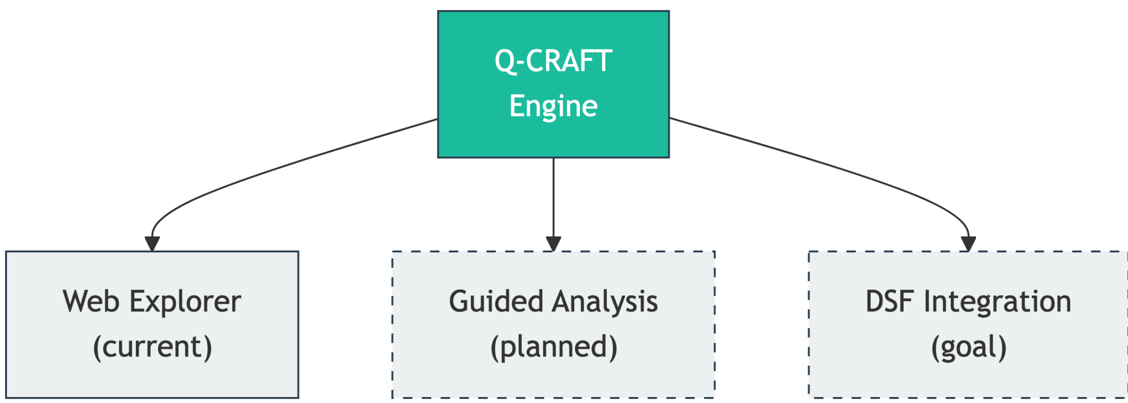
The testing pipeline is public and automated. Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation across debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP, with a further 25 parameter sensitivity combinations passing. Climate-scenario parity is confirmed for ratio metrics, meaning debt-to-GDP, revenue and expenditure as percent of GDP. This kind of transparent, automated verification is essential for tools where the stakes are higher. Chapter 2 has the fuller statement of what the claim covers.

A.3 Q-CRAFT is the proof of concept

Q-CRAFT is deliberately small: five user-facing parameters, one core equation. We chose it because it demonstrates the approach in miniature.

The real goal is the LIC-DSF, the Low-Income Country Debt Sustainability Framework. The LIC-DSF modeling chain (Q-CRAFT, DIGNAD, LIC-DSF) is far more complex, far more consequential, and used for every low-income country. We are already working with end-users of the LIC-DSF, and we want to work with its designers too.

A.4 A modular engine, multiple interfaces



One engine, multiple interfaces

The same engine powers different front-ends for different needs:

1. **Web Explorer (current).** Interactive charts for quick country analysis. Select a country, adjust parameters, compare scenarios.
2. **Guided Analysis Tool (planned).** AI-assisted reasoning through input assumptions. Always available, cheaper than a technical assistance mission. Helps users understand what to enter and why.
3. **DSF Integration (the goal).** Q-CRAFT and DIGNAD feeding into LIC-DSF assumptions, with the same ergonomic layer applied to a far more consequential tool.

The current process makes it hard to do things the right way. Documentation is separated from implementation. If you do not know what to enter for a parameter, you search the user guide and hope it is there (and often it is not). The goal is design that makes it easy to do the right thing and hard to do the wrong thing.

A.5 The co-design invitation

We built V1. Here are specific areas where practitioner input shapes V2.

1. **Parameter guidance.** How should we help users choose productivity growth, expenditure rigidity, and debt target values? What heuristics do experienced practitioners use?
2. **Default assumptions.** What are sensible defaults by country type? Should defaults vary by region?
3. **Capacity development integration.** How would this tool fit into an actual CD workshop or IMF mission?
4. **Validation priorities.** Which countries and scenarios should we prioritize for verification?
5. **Missing features.** What is the first thing you would want that is not here?

What we are asking for: a few hours from a few people for design feedback, continued communication so we build credibly, and feedback on whether we are making something useful.

What we are NOT asking for: official endorsement (we understand institutional constraints), access to confidential information (we work with publicly available materials), or money (this is grant-funded, MIT-licensed open source).

A.6 Why start here

Two reasons to start with Q-CRAFT:

1. **Immediate value.** A co-designed Q-CRAFT tool could be in use within weeks. It costs nothing and requires minimal commitment to evaluate. It complements, not replaces, the Excel workbook.
2. **Pioneer the approach.** Success with Q-CRAFT builds trust and methodology for the LIC-DSF. We prove the design process works on a small tool before applying it to the one that matters most.

A.7 The SovTech vision

Central banks have been applying technology to supervisory functions for years. The BIS documented "SupTech generations" as a framework for technology-driven supervisory and regulatory solutions. The question is no longer whether such tools belong in public financial institutions. The question is when sovereign debt management will catch up.

We call this **SovTech**: the application of SupTech principles to sovereign debt analysis. Three principles guide this work:

Modularity that respects how people work. Same engine, different interfaces. Components like building blocks: swap one module without rebuilding everything.

Open source as institutional infrastructure. Transparency means anyone can examine how analysis is done. MIT license, no barriers to adoption. The calculation engine and the interface are separate concerns, developed and tested independently.

AI as enabler, with a trust layer. The cost of building analytical software has dropped dramatically. But sovereign debt analysis serves policy decisions worth billions of dollars. Verification pipelines, automated testing, and human-in-the-loop governance are requirements, not optional features. This is why we built the parity verification pipeline described in Chapter 2.

Q-CRAFT Explorer is our attempt to show what this looks like. The source code is on GitHub. We welcome your input.

Glossary

Climate scenarios Six warming pathways used in Q-CRAFT, based on IPCC Shared Socioeconomic Pathways (SSPs). Range from Paris-aligned (below 2°C warming by 2100, SSP1-2.6) to Hot Unadapted (high warming with slow adaptation, SSP3-7.0 at the 90th percentile). Climate impacts on GDP are derived from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on the empirical estimates of Kahn et al. (2021). See the IMF User Guide, Section IV.B for detailed methodology.

C-PIMA (Climate Public Investment Management Assessment) An IMF framework for evaluating how well a country's public investment management institutions account for climate change. C-PIMA assessments include an annex that uses Q-CRAFT to model the fiscal impacts of climate scenarios for the assessed country. See the C-PIMA Handbook (IMF, 2025) for the full framework.

DSA (Debt Sustainability Analysis) A framework used by the IMF and World Bank to assess whether a country's debt is sustainable (that is, whether the government can meet its current and future debt obligations without requiring debt relief or accumulating arrears). Q-CRAFT projections can complement DSA analysis by providing long-term climate-adjusted fiscal trajectories.

Debt dynamics equation The core equation in Q-CRAFT: $d_t = d_{t-1} \times \frac{1+r_t}{1+g_t} - pb_t$. Next year's debt-to-GDP ratio equals the current ratio, adjusted for the interest-growth differential, minus the primary balance. When interest rates exceed growth ($r > g$), debt rises automatically: the government must run a primary surplus just to keep debt stable. See the IMF User Guide, Section IV.A on debt dynamics for detailed derivation and corollaries.

Debt-to-GDP ratio Total government debt expressed as a percentage of nominal GDP. The standard measure of a country's debt burden, since it scales debt relative to the economy's ability to service it. Q-CRAFT projects this ratio through 2099 under baseline and climate scenarios.

Expenditure rigidity The degree to which government spending resists downward adjustment when climate shocks reduce GDP. Measured on a 0-1 scale. A value of 1.0 means spending is fully sticky: it stays at the baseline level in local currency terms even as GDP falls, representing the worst case for debt accumulation. A value of 0.0 means spending adjusts fully with GDP, maintaining the baseline expenditure-to-GDP ratio. This parameter only affects climate scenarios, not the baseline. See the IMF User Guide, pp. 20 and 35-36.

Fiscal rule A mechanism in Q-CRAFT that adjusts primary expenditure to steer debt toward a target ratio. When enabled, the model computes the debt-stabilizing primary balance and adjusts spending by the fiscal gap (the difference between the actual and debt-stabilizing primary balance). The adjustment is additive in levels (not a rate) and depends on the prior year's state, requiring recursive year-by-year computation. See the IMF User Guide, pp. 15-18.

DIGNAD (Debt, Investment, Growth, and Natural Disasters) An IMF dynamic general equilibrium model for analyzing the macroeconomic effects of natural disasters and public investment in developing countries. More complex than Q-CRAFT, with substantially more parameters. Discussed in the co-design appendix as an example of a tool that could benefit from the SovTech approach.

Golden master In software verification, a reference output used to detect unintended changes. Q-CRAFT's golden master tests compare the Python engine's outputs against values extracted from the IMF's Excel workbook for the same country and parameter inputs. Parity with the Excel tool (not any independent calculation) is the acceptance criterion.

LIC-DSF (Low-Income Country Debt Sustainability Framework) The IMF-World Bank framework for assessing debt sustainability in low-income countries. Uses country-specific debt burden thresholds based on a composite indicator of institutional quality and macroeconomic fundamentals. Compared against Q-CRAFT in Module 5, and discussed in the co-design appendix as a more complex tool that could benefit from the SovTech approach.

Interest-growth differential The difference between the nominal interest rate on government debt (r) and nominal GDP growth (g), expressed as $(r - g)/(1 + g)$. When positive ($r > g$), debt dynamics are unfavorable: the debt ratio rises automatically. When negative ($g > r$), debt dynamics are favorable. This single variable determines whether debt is self-stabilizing or self-reinforcing.

Primary balance Government revenue minus non-interest expenditure (primary expenditure), expressed as a share of GDP. The primary balance is the fiscal variable that a government can control in the near to medium term through fiscal policy. A primary surplus means the government generates enough revenue to cover its non-interest spending; a primary deficit means it does not.

WEO (World Economic Outlook) The IMF's flagship publication providing analysis and projections of the global economy. Q-CRAFT uses the WEO database (currently October 2024 vintage) as the source for macroeconomic and fiscal data through the medium-term forecast horizon (currently through 2029). Beyond this horizon, Q-CRAFT's own projection methodology takes over.

SovTech The application of SupTech (supervisory technology) principles (modularity, open source, human-centered design) to sovereign debt analysis and public financial management tools. Q-CRAFT Explorer is a proof of concept for this approach.

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